# Architecture and Design Document

*Software Design II - Universidad del Norte | Project 5 / RFP DH-05 - Internship and Practicum Management System*

*Evaluation 2 - Architecture: justifying the solution structure | Team: Sebastián Mercado - Cher Jiménez - Juan Arbeláez*

Evaluation 1 answered 'do we understand the problem?'. This document answers the central question of Evaluation 2: can the team justify how the system should be structured? It does not start from technology. It starts from the evidence of Evaluation 1, turns the quality requirements into measurable scenarios, selects the few facts that drive the structure, compares alternatives, records the decisions, and only then shows the views, the data, the interactions and their failures, the contracts, the security model, the validation with its honest status, and the work that remains.

Every claim carries one of four labels. DESIGNED: decided and documented, no code. IMPLEMENTED: code exists. VERIFIED: code exists and an automated test that we ran exercises it. NOT EXECUTED: planned and not run, with the reason. Two qualifiers appear in the test plan: EXECUTED marks a check that was run but does not verify a feature by itself (the mutation check), and PARTIAL marks a test that verifies only part of what it targets. A diagram or a contract specifies behavior; it does not prove that the behavior exists.

| Companion artifact | What it holds |
|---|---|
| Architecture Decision Records - DH-05 | ADR-000 to ADR-010 with status, date, owner, context, decision, alternatives, consequences, verification, reconsideration triggers and trace links. |
| Artifacts architecture drivers and decisions.xlsx | Every table of this document as a sheet (27 sheets), in the format of the Evaluation 1 workbooks. |
| dh05-internship-system (repository) | README as evidence index; docs/evaluation-2 with views, ADRs, design checks, test plan, validation evidence and project status; the implemented slice and its tests. |

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# 1. Purpose and agreed scope

## Problem

Horizon University coordinates 600 to 900 placements a year with e-mail, spreadsheets, forms and shared folders. Nobody can say reliably in which state a placement is, a student can start before every institutional condition is met, host organizations see more student data than they need, and a placement can be closed while the academic evidence is still missing (RFP sections 1 and 2; Evaluation 1).

**Desired outcome.** One controlled lifecycle from opportunity to closure, gates that cannot be bypassed at activation and closure, access that follows the relationship with the placement, and a history of every decision that nobody can rewrite. Not a nicer spreadsheet.

## What we take from Evaluation 1

| Evaluation 1 artifact | How Evaluation 2 uses it |
|---|---|
| Quality Requirements QR-01 to QR-05 | Source of the quality scenarios and of four of the five drivers. |
| Risk Register R-001 to R-010 | Risks that became drivers: rules changing (R-001), early start (R-002), over-exposure (R-003, R-004), failing services (R-005), copying the export as schema (R-008). |
| Clarification and Assumption logs | Open questions the design must survive (the seven of section 15: CL-001, CL-004, CL-008, CL-016, CL-017, CL-018, CL-020) and the assumptions about authoritative services (ASM-002, ASM-003, ASM-009). |
| Process Analysis OB-01 to OB-12, 15 stages | The stages give the module boundaries; OB-07, OB-11 and OB-12 give the gates and the history. |
| Capabilities CAP-01 to CAP-06 and stories US-001 to US-030 | The significant requirements of section 2; every story has a home module (section 14). |
| Success Criteria SC-01 to SC-12 | The verification column of the traceability and the acceptance tests of the slice. |

Evaluation 1 is the input of this evaluation and its artifacts stay as they were delivered. While designing, we read the data pack and the starter kit in more detail and recorded the facts that refine what we now build on (evidence updates EU-01 to EU-05):

| ID | What the evidence says | How we use it |
|---|---|---|
| EU-01 | data/reference/placement_types.csv defines ten active types (ACADEMIC_INTERNSHIP, PROFESSIONAL_PRACTICUM, COOPERATIVE_ EDUCATION, RESEARCH_INTERNSHIP, COMMUNITY_ SERVICE, TEACHING_PRACTICUM, CLINICAL_FIELD, | For the architecture we take the file as the source: ten defined types, five used by opportunities, three by placements. The Evaluation 1 artifacts stay as delivered. The question behind CL-001 is unchanged and still open (which types enter the first release is a |

| ID | What the evidence says | How we use it |
|---|---|---|
| | INTERNAL_UNIVERSITY, INTERNATIONAL, REMOTE_ INTERNSHIP). The records use fewer: internship_ opportunities.csv uses five of them (academic internship, professional practicum, community service, research internship, remote internship) and placements.csv uses three. | client decision), and the architecture does not depend on the number: placement types are reference data and rules are configured per type (ADR-004). |
| EU-02 | Two PROFESSIONAL_PRACTICUM placements have different checklists: PLC-2001 has ACADEMIC_ APPROVAL, VALID_AGREEMENT and WORK_PLAN; PLC-2003 has VALID_AGREEMENT and INSURANCE_ REFERENCE. | The inconsistency confirms that the checklist cannot be fixed in code. The initial configuration uses the union of the observed conditions and is marked as pending CL-020. |
| EU-03 | students_current_export.csv uses the legacy code SWE for HU-STU-5006, while academic_programs.csv lists it as an alias of BSC-SWE. placements.csv stores 'Blocked' as a status, although blocking is a consequence of the readiness checklist. | Program codes are normalized through known_aliases during import, and 'Blocked' is not a lifecycle state: it is derived from the readiness gate. Both transformations are written in data/generated/transformation_log.md. |
| EU-04 | The starter reads data/reference/mock_users.json and exposes /mock/institutional/students, but the DH-05 pack ships mock_identities.json and a contract for /mock/student-information/students/{id}. The starter has no Employer Directory mock. | We keep the starter and write adapters for the DH-05 contracts, including a new Employer Directory mock with its three failure responses (ADR-000, ADR-007). |
| EU-05 | mock_identities.json contains no workplace-supervisor identities (no SUP-EXT users), so the API answers 401 for them before the access policy is reached. | QS-02 is verified at policy level (test_qs02_workplace_ supervisor_is_related_only_to_their_own_placement) and through the API with an unrelated student (test_ qs02_unrelated_student_gets_404_and_denial_is_ audited). The end-to-end case with a real supervisor identity waits for the identity service or a fixture identity (BL-06). |

## Scope

In scope for the product:

- Opportunity registration and publication review (stages 1-3).
- Applications, eligibility decision, employer decision and academic approval as three separate results (stages 4-7).
- Placement lifecycle, work plan versions, supervisor assignment, readiness checklist, activation gate (stages 7-12).
- Hours in three levels, incidents, evaluations, closure gate (stages 13-15).
- Role and relationship access with audience projections, append-only audit history, notifications through an outbox, operational reports.

Out of scope:

- Payroll, stipends or any real financial transaction (RFP section 10).
- Automatic candidate selection or eligibility decided by artificial intelligence (RFP section 10).
- Real integrations with the identity, student-information, employer-directory and e-mail systems: they are simulated (RFP section 7).
- Storing the documents themselves: DH-05 stores evidence references only.
- Web client technology and screens: decided in Evaluation 3 because no driver depends on them.

Implemented in this evaluation as a foundation, not as the product: the activation gate (US-014, US-016) with access, audit and outbox. The other modules are designed and appear in the structure; they are built in Evaluation 3.

## Stakeholders become concerns and design consequences

| Stakeholder | Concern | Design consequence |
|---|---|---|
| Student (STK-01) | Know what is missing and when; see only feedback meant for them | Readiness projection per audience; restricted fields never leave M-07 (ADR-005). |
| Host organization (STK-02) | Review candidates without bureaucracy | Employer projection of an application; no academic profile (QS-01). |
| Workplace supervisor (STK-03) | Confirm hours and evaluate | Relationship-based access to assigned placements only (QS-02). |
| Faculty supervisor (STK-04) | See progress and evidence before the signoff | Closure gate owned by M-03 with evidence from M-05 (BR-06). |
| Academic program coordinator (STK-05) | Rules applied the same way for the whole program | Versioned rules evaluated by Strategy; the version is stored in each decision (ADR-004). |
| Experiential-learning coordinator (STK-06) | Never activate a placement that is not ready | Readiness gate in one transaction with a conditional update (ADR-003, ADR-010). |
| Legal representative (STK-07) | No activation with an expired agreement | Agreement verified in the Employer Directory at activation; fail closed (ADR-007). |
| IT Services (STK-08) | Maintainable, controlled integrations, small team | Modular monolith, ports and adapters, architecture test in the test suite (ADR-001, ADR-007). |
| Office director (STK-09) | One process, visible risk | Audit history and at-risk report (M-08, M-10). |
| Data privacy officer (STK-10) | Minimum data to external users | Deny by default while CL-018 is open; threat model T-02..T-04. |

# 2. Significant requirements, rules and assumptions

## Vocabulary used in every view

| Term | Meaning |
|---|---|
| Opportunity | A role posted by a host organization and reviewed before students see it. |
| Application | A student's request for one opportunity, carrying three independent results. |
| Placement | The internship itself, created when a student is selected; the aggregate root of the lifecycle. |
| Readiness condition | One row of the checklist that must be COMPLETE and not expired before activation. |
| Readiness gate | The evaluation of every required condition for a placement under one rule version. |
| Activation | The committed change of a placement to ACTIVE. It is a decision, not the first working day. |
| Start date | The day the student really begins; recorded after activation. |
| Rule version | The identifier of the rule configuration used for one evaluation (readiness-v1). |
| Audit entry | An append-only record of one decision or denial with actor, values, reason and correlation id. |
| Outbox message | A notification written with the decision and delivered later by the worker. |
| Correlation id | One identifier per command, copied to its audit entry, event and messages. |

Four distinctions that the whole design protects:

- Selected by the employer is not academically approved.
- Ready (the gate passes) is not active (the command committed).
- Active is not started (the start date is recorded later).
- Activated is not notified (delivery is eventual).

## Significant requirements

We do not copy the thirty stories. These are the Evaluation 1 capabilities and cross-cutting stories that have architectural consequences, with the status of each one today.

| ID | Requirement | Stories | Architectural significance | Status |
|---|---|---|---|---|
| CAP-01 | Controlled publication with four review results | US-001..US-004 | Opportunity lifecycle owned by M-01; review result audited. | DESIGNED |
| CAP-02 | Eligibility checked at application time | US-005..US-008 | Versioned rules (M-06) plus Student Information port with fail-soft policy (QS-05). | DESIGNED |
| CAP-03 | Three approvals kept apart | US-009..US-011 | Three separate fields and events; none implies another (BR-05). | DESIGNED |
| CAP-04 | Readiness gate before | US-012..US-016 | Gate + transition table + conditional update in | VERIFIED (slice) |

| ID | Requirement | Stories | Architectural significance | Status |
|---|---|---|---|---|
| | activation | | one transaction (AD-01). | |
| CAP-05 | Verified hours and evidence | US-017..US-021 | Three levels of hours in M-05; dispute role configurable (CL-017). | DESIGNED |
| CAP-06 | Evidence-based closure and history | US-022..US-026 | Closure gate (BR-06) and append-only audit (ADR-006). | Audit VERIFIED; closure DESIGNED |
| US-027, US-028 | Identity and relationship-based access | US-027, US-028 | One access module, audience projections, deny by default (ADR-005). | VERIFIED (slice) |
| US-030 | Notifications that never undo a decision | US-030 | Transactional outbox and worker (ADR-008). | VERIFIED |

## Business rules and invariants that the design must enforce

| ID | Rule | Source | Enforced in | Verification |
|---|---|---|---|---|
| BR-01 | A placement becomes ACTIVE only when every required readiness condition is COMPLETE and not expired, the agreement is valid in the Employer Directory, and both supervisors are assigned. | RFP section 7; OB-07; ASM-006; R-002 | M-03 + M-04 (ADR-003) | test_sc01_*, test_ sc02_*, test_ready_ placement_is_ activated_*, tests/unit/ test_readiness_rules.py |
| BR-02 | If the agreement cannot be verified because the directory fails, the activation is refused (fail closed). | ASM-003; R-005; QS-08 | M-04 AgreementValid InDirectory (ADR-007) | test_directory_down_ means_activation_ refused, test_directory_ unavailable_keeps_gate_ closed |
| BR-03 | Activation and the real start date are different facts: the start date is recorded after activation and is not a status. | OB-07 | M-03 Placement. register_start (ADR-003) | test_start_date_only_ after_activation |
| BR-04 | 'Blocked' is not a lifecycle state; it is the result of evaluating the gate. | EU-03; R-008 | M-03 transition table, seed STATUS_MAP | test_active_cannot_be_ reached_by_other_ commands; transformation_log.md |
| BR-05 | Eligibility, employer decision and academic approval are three separate results; none implies another. | OB-05; CL-003 | M-02 (designed) | NOT EXECUTED - module planned for Evaluation 3 |
| BR-06 | A placement is completed only when the closure gate passes: accepted hours reach the required total, final evaluations are submitted without hold and the academic signoff exists. | OB-11; R-007; DH05-SC-05 | M-03 + M-05 (ADR-003) | NOT EXECUTED - designed; closure gate is Evaluation 3 |
| BR-07 | Every state-changing command and every denied access writes an audit entry; entries are never updated or deleted. | QR-03; OB-12; US-026 | M-08 (ADR-006) | test_audit_history_can_ only_be_appended_ and_read, audit assertions in activation tests |
| BR-08 | Every eligibility or readiness decision stores the | QR-05; F-24; QS- | M-06 (ADR-004) | test_rule_change_is_ |

| ID | Rule | Source | Enforced in | Verification |
|---|---|---|---|---|
| | version of the rules used; changing a rule never changes a past decision. | 07 | | configuration_only, test_qs04_audit_reason_records_the_rule_version_used |
| BR-09 | Restricted evaluation comments and RESTRICTED/INTERNAL incidents are visible only to authorized roles; while CL-018 is open, access is denied by default. | QR-02; CL-018; R-004 | M-07 (ADR-005) | test_host_organization_sees_readiness_without_restricted_evidence (evidence references); evaluation/incident projections NOT EXECUTED (M-05 planned) |
| BR-10 | A user reaches a placement only through a relationship with it; without one, the answer is 'not found'. | QR-02; ASM-007 | M-07 (ADR-005) | test_unrelated_user_gets_not_found, test_other_organization_cannot_see_the_placement, test_qs02_* |
| BR-11 | A failed notification never cancels or reverts an approval, an activation or a closure. | QR-04; ASM-009 | M-09 outbox (ADR-008) | test_failed_delivery_keeps_activation_and_retries |
| BR-12 | Two commands on the same placement cannot both succeed from the same starting status. | Design rule found in review (QS-09) | M-03 repository conditional update (ADR-010) | test_two_concurrent_activations_only_one_commits |

## Constraints

| ID | Constraint | Source |
|---|---|---|
| C-01 | Team of three students and one semester. | RFP section 7; course |
| C-02 | Only synthetic data; integrations with identity, student, document and academic systems may be simulated. | RFP section 7; ASM-011 |
| C-03 | The university has limited technical personnel and prefers maintainable solutions. | RFP section 7; STK-08 |
| C-04 | The solution must operate independently of the other student projects. | RFP section 7 |
| C-05 | No payroll or real financial transactions; no automatic candidate selection with artificial intelligence. | RFP sections 7 and 10 |
| C-06 | A technical starter kit (Python, FastAPI, SQLAlchemy, SQLite) is supplied and may be replaced if the decision is documented. | horizon-common-starter README |

## Design assumptions

Evaluation 1 recorded twelve assumptions about the problem. These six are about the design; each says how we would find out that it is false and what would change. None is client-confirmed.

| ID | Assumption | Source | How to validate | If false |
|---|---|---|---|---|
| DA-01 | Load stays between 600 and 900 placements per year with 8 to 12 office users. | RFP section 8; F-22 | Ask the office for last year's numbers (CL-005 follow-up). | ADR-001 revisit triggers: measure and reconsider the structure. |
| DA-02 | One institution, one time zone; all timestamps are stored in UTC. | Starter kit convention | Confirm no campus works in another country. | Add a time-zone attribute per campus; no structural change. |
| DA-03 | SQLite is enough for local development and the demo; PostgreSQL is the shared environment. | Starter docker-compose | Run the same tests against PostgreSQL (NOT EXECUTED, see TP-14). | Fix SQL differences; the repository isolates them. |
| DA-04 | The mock identity header is a development mechanism, never production security. | Starter technical assumptions | Decide with IT Services the real authentication before any real pilot. | Only IdentityAdapter changes (ADR-007). |
| DA-05 | One experiential-learning coordinator role is allowed to activate placements; other roles are not. | US-016; CL-008 open | Confirm with the office which staff roles may activate. | Change one entry of the access policy (M-07). |
| DA-06 | The Employer Directory answers the agreement question for the organization at the moment of activation; DH-05 does not cache the answer between requests. | ASM-003 | Contract test against the real service when it exists. | Add a snapshot with timestamp (already designed for verification evidence). |

## One synthetic scenario used throughout

Every view, sequence, test and screen uses the same records, so the reader can follow one placement across the whole document.

| Record | What it holds | Where it is used |
|---|---|---|
| PLC-2002 | Selected by ORG-001, academic approval READY-004 and work plan READY-006 pending | Refused activation (SEQ-02, DH05-SC-01) |
| PLC-2003 | Agreement READY-007 expired, insurance READY-008 missing; ORG-005 agreement EXPIRED in the directory | Refused activation (DH05-SC-02) |
| PLC-9001 | Synthetic test placement (not in the data pack): ORG-001, all conditions complete | Successful activation, concurrency, rollback (SEQ-01, SEQ-04) |
| PLC-2001 | Already ACTIVE in the export, student HU-STU-5003. Its readiness read shows not ready under readiness-v1 (INSURANCE_REFERENCE not registered): a consequence of EU-02, not a defect. Its activated_at (2027-01-20) is later than the fixed test date (2026-09- | Seed exception of QS-08; unrelated-user tests |

| Record | What it holds | Where it is used |
|---|---|---|
| | 24) because the synthetic pack describes a future period; we load it as exported and no rule uses that date | |
| EXP-COORD-001 | Experiential-learning coordinator (staff role) | The only role allowed to activate |
| HU-STU-5001 / HU-STU-5003 | Students of PLC-2002 / PLC-2001 | Denied activation; not-found for unrelated placement |
| ORG-C-001 | Contact of ORG-001 | Host-organization projection; 404 for PLC-2003 |
| Fixed date | 2026-09-24 in tests | Agreement validity does not change with the calendar |

### Use case specification of the critical path

| | |
|---|---|
| **Use case** | UC-ACT  Activate a placement (US-014, US-016) |
| **Primary actor** | Experiential-learning coordinator (staff role EXPERIENTIAL_LEARNING_COORDINATOR). |
| **Trigger** | The coordinator selects Activate on a placement in PENDING_ACTIVATION. |
| **Preconditions** | Valid identity; the placement exists; the actor's staff role allows the action. |
| **Main flow** | Resolve identity, check access, read the placement, evaluate the gate under the current rule version, transition to ACTIVE, save with a conditional update, write the audit entry, publish PlacementActivated (outbox), commit, return status, rule version and correlation id. |
| **Success guarantee** | One committed ACTIVE status with activated_at, exactly one PLACEMENT_ACTIVATED audit entry and three outbox messages, all with the same correlation id. |
| **Failure guarantee** | No partial state: the placement stays PENDING_ACTIVATION, no success audit entry, no messages. |
| **Alternatives** | Gate blocks (409 ACTIVATION_BLOCKED, audited); directory down (blocks, fail closed); unrelated user (404, audited); related but not allowed (403, audited); concurrent command first (409 STATE_CHANGED); already active (409 ILLEGAL_TRANSITION); identity disabled (503). |
| **Visible result** | The office sees ACTIVE immediately; notifications may still be pending. |
| **Status** | IMPLEMENTED and VERIFIED by TP-04 and TP-05. |

# 3. Quality attribute scenarios

A quality word is not useful for design until it is measurable. Each scenario follows the six-part form (source, stimulus, environment, artifact, response, measure) and uses a real record of the data pack or a documented failure of the simulated services, so that it can become a test. The

measures are targets for the system we build, not measurements of the current process. QS-09 was added after reviewing the first slice.

## QS-01 - Privacy (QR-01)

| | |
|---|---|
| **Source** | Host-organization contact ORG-C-001 (Northbridge Digital Services), authenticated as an external user. |
| **Stimulus** | Opens the candidate INT-APP-1001 (student HU-STU-5001) to take the hiring decision. |
| **Environment** | Normal operation, employer-review stage (stage 6). |
| **Artifact** | Applications module, candidate view. |
| **Response** | The system returns only the candidate projection approved for employers: name, program, the documents of that application classified as APPLICATION (CV, motivation letter) and the application status. Academic average, credits, standing, other applications of the student and RESTRICTED documents are never included. |
| **Measure** | 0 fields outside the employer projection in 100% of responses to external users. Verified by an automated test per external endpoint. |
| **Evidence** | CL-009, ASM-007, document_types.csv (sensitive_level), SC-08 |
| **Priority** | Business importance H / architectural impact H |

## QS-02 - Relationship-based Authorization (QR-02)

| | |
|---|---|
| **Source** | Workplace supervisor SUP-EXT-002 (ORG-001), authenticated and with a valid supervisor role. |
| **Stimulus** | Requests the evaluations and incidents of placement PLC-2001, which belongs to ORG-002 and is supervised by SUP-EXT-001. |
| **Environment** | Normal operation. |
| **Artifact** | Access module, policy decision for every placement query. |
| **Response** | The request is denied because the user has no relationship with PLC-2001, even though their role is correct. The response does not reveal whether the placement exists, and the attempt is written to the audit history. |
| **Measure** | 100% of requests without a relationship are denied; 100% of denials are recorded with actor, resource and time. |
| **Evidence** | ASM-007, CL-018, US-028, SC-08, SC-09 |
| **Priority** | Business importance H / architectural impact H |

## QS-03 - Field-level privacy (QR-01, QR-02)

| | |
|---|---|
| **Source** | Student HU-STU-5003 (placement PLC-2001). |
| **Stimulus** | Opens the midterm evaluation EVAL-001 and the incident list of their own placement. |
| **Environment** | Normal operation, monitoring stage (stage 13). |
| **Artifact** | Monitoring & Evaluation module, evaluation and incident projections. |
| **Response** | The student sees student_visible_feedback and the rating, but never restricted_comments. Incidents with visibility RESTRICTED or INTERNAL (for example INC-001 and INC-002 in the data pack) are shown only to the roles allowed for that visibility. While CL-018 is open, restricted fields are denied to everybody except the experiential-learning office, the faculty supervisor of the placement and the author of the content (a workplace supervisor reads their own comments); deny by default. |
| **Measure** | restricted_comments absent from 100% of student and external responses; tested for every audience in the access matrix. |
| **Evidence** | evaluations.csv (MIXED visibility), placement_incidents.csv (visibility), CL-018, R-004, SC-07, SC-09 |
| **Priority** | Business importance H / architectural impact M |

## QS-04 - Auditability (QR-03)

| | |
|---|---|
| **Source** | Experiential-learning coordinator EXP-COORD-001 or an auditor of the program. |
| **Stimulus** | Investigates months later why hour entry HOUR-002 was reported as 10 hours and confirmed as 8, and which work plan version was approved for PLC-2001. |
| **Environment** | Normal operation, after the placement is closed. |
| **Artifact** | Audit History module and the versioned records (work plan, decisions, hours). |
| **Response** | The history of the placement shows every change with actor, date and time, previous value, new value and reason. WP-001 is labeled v2, so the previous approved version and v2 both remain readable; nothing was overwritten. |
| **Measure** | 100% of state-changing commands write an audit entry in the same database transaction as the change. No application interface can update or delete an audit entry. The history of one placement is obtained with one query. |
| **Evidence** | OB-12, F-02, DH05-SC-03, DH05-SC-04, US-026, US-021, SC-10 |
| **Priority** | Business importance H / architectural impact H |

## QS-05 - Reliability (integration failure) (QR-04)

| | |
|---|---|
| **Source** | Simulated Student Information Service. |

| | |
|---|---|
| **Stimulus** | Answers SIS-503 SERVICE_UNAVAILABLE (or SIS-409 STALE_ACADEMIC_PROFILE) while the eligibility of a new application is checked. |
| **Environment** | Degraded operation, peak at the beginning of the academic period (RFP section 8). |
| **Artifact** | Applications module and the Student Information adapter. |
| **Response** | The application is saved. Its eligibility stays ELIGIBILITY_PENDING with the error code recorded, and the check is retried. A 409 answer sends the case to manual review. The system never decides eligibility without a source timestamp. |
| **Measure** | 0 applications lost; 0 eligibility decisions without source data and timestamp; the student sees the pending status and the reason. |
| **Evidence** | student_information_service/failure-responses.json, ASM-002, ASM-009, R-005, eligibility_reviews.csv (source_data_timestamp), SC-11 |
| **Priority** | Business importance H / architectural impact M |

## QS-06 - Reliability (notification failure) (QR-04)

| | |
|---|---|
| **Source** | Simulated notification channel. |
| **Stimulus** | Fails when the system notifies the student and the supervisors that PLC-2001 was activated. |
| **Environment** | Normal operation. |
| **Artifact** | Notifications module (outbox). |
| **Response** | The activation stays committed. The notification is kept in the outbox with its failure and is retried; the failure is visible to the office. |
| **Measure** | 0 approvals, activations or closures rolled back because of a notification failure; every failed notification is retried or flagged. |
| **Evidence** | ASM-009, US-030, SC-11 |
| **Priority** | Business importance M / architectural impact M |

## QS-07 - Maintainability (rule change) (QR-05)

| | |
|---|---|
| **Source** | Academic Program Coordinator ACAD-COORD-101, through the experiential-learning office. |
| **Stimulus** | The BSc Biology program adds CONFIDENTIALITY_FORM as a readiness condition for PROFESSIONAL_PRACTICUM, and raises its minimum average, while applications are open. |
| **Environment** | Normal operation, during an open application period. |
| **Artifact** | Rule Configuration module. |

| | |
|---|---|
| **Response** | The change is applied as a new version of the rule configuration. New evaluations use the new version; decisions already taken keep the version they were evaluated with. |
| **Measure** | No source-code change and no redeployment; 0 past decisions modified; the rule version is visible in every eligibility decision and readiness evaluation. |
| **Evidence** | RFP section 7, F-24, CL-004, CL-020, ASM-004, R-001, EU-02 |
| **Priority** | Business importance H / architectural impact H |

## QS-08 - Integrity of the activation gate (RFP section 7 constraint; supports QR-03, QR-04)

| | |
|---|---|
| **Source** | Experiential-learning coordinator EXP-COORD-001. |
| **Stimulus** | Tries to activate PLC-2002 (academic approval and work plan pending) and PLC-2003 (agreement expired, insurance missing). |
| **Environment** | Normal operation; also when the Employer Directory Service is unavailable. |
| **Artifact** | Placements module, activation command and readiness gate. |
| **Response** | Both activations are refused and the system names the blocking conditions: READY-004 and READY-006 for PLC-2002; for PLC-2003, READY-007 (expired), READY-008 (missing), the directory's EXPIRED agreement for ORG-005 (valid until 2026-07-31), and ACADEMIC_APPROVAL and WORK_PLAN, which the configuration requires but its checklist does not contain (EU-02). If the agreement cannot be verified because the directory is down, the activation is also refused (fail closed). The attempt is recorded. |
| **Measure** | 100% of activations with a pending, blocked, expired or unverifiable required condition are refused; after the initial load there is no other path (import, API, script) that sets a placement to ACTIVE. The one exception is the one-time seed, which loads historic placements as exported (PLC-2001 is already ACTIVE in the data pack) and logs it in data/generated/transformation_log.md. |
| **Evidence** | RFP section 7, ASM-006, R-002, DH05-SC-01, DH05-SC-02, SC-05 |
| **Priority** | Business importance H / architectural impact H |

## QS-09 - Consistency under concurrent commands (RFP section 7 constraint; supports QR-03)

| | |
|---|---|
| **Source** | Two experiential-learning coordinators, or one coordinator who double-clicks. |
| **Stimulus** | Both send the activation command for the same placement in PENDING_ACTIVATION at the same moment, and both read the same status before either writes. |
| **Environment** | Normal operation. |
| **Artifact** | Placements module: repository and activation command. |

| | |
|---|---|
| **Response** | Exactly one command writes ACTIVE. The other rolls back, receives 409 with code STATE_CHANGED and leaves no success audit entry and no notification. |
| **Measure** | For N concurrent commands on the same status, exactly 1 success audit entry and 1 set of notifications; N-1 receive STATE_CHANGED or ILLEGAL_TRANSITION. Verified by a deterministic interleaving on SQLite; true parallel run on PostgreSQL NOT EXECUTED (TP-14). |
| **Evidence** | RFP section 7, AD-01, review of the first slice (ADR-010), DH05-SC-01 |
| **Priority** | Business importance M / architectural impact H |

Some qualities were deliberately not turned into drivers:

- **Performance and scalability:** RFP section 8 and F-22: 600 to 900 placements per year, 8 to 12 office users and a seasonal peak at the start of each academic period. This load is small for any conventional web application, so no scenario requires special scaling decisions. We keep a revisit trigger in ADR-001.
- **24/7 high availability:** No stakeholder asked for continuous availability, and the RFP only requires that failures do not lose data (QR-04). Reliability is treated as data consistency, not as uptime.
- **Interoperability with production systems:** RFP section 7 allows simulated integrations. We protect the boundary with adapters (ADR-007) instead of designing for real institutional APIs.

# 4. Architecture drivers and tensions

From 30 stories, 10 stakeholders, 20 clarifications, 12 assumptions, 10 risks and 12 success criteria, five facts decide the structure. A driver is a fact that, if it changed, would change the architecture.

| ID | Driver | Comes from | Architectural consequence | Decisions |
|---|---|---|---|---|
| AD-01 | Activation and closure gates that cannot be bypassed | RFP section 7, OB-07, OB-11, ASM-006, R-002, R-007, QS-08, QS-09 | Placement lifecycle changes happen only through commands of the Placements module, which evaluates the readiness gate or the closure gate in the same transaction. No other module, import or script writes the placement status. | ADR-001, ADR-002, ADR-003, ADR-010 |
| AD-02 | Access by relationship and by field | QR-01, QR-02, CL-009, CL-018, ASM-007, R-003, R-004, QS-01, QS-02, QS-03 | One Access module decides every request (deny by default) using role plus relationship with the opportunity or placement, and every response goes through an audience projection that removes the fields the user may not read. | ADR-005 |
| AD-03 | Complete history that | QR-03, OB-12, F-02, US-026, | Every command writes an append-only audit | ADR-002, ADR- |

| ID | Driver | Comes from | Architectural consequence | Decisions |
|---|---|---|---|---|
| | nobody can rewrite | CL-019, QS-04, QS-09 | entry in the same transaction. Work plans, eligibility decisions and hour reviews are stored as new records or versions, never overwritten. | 006, ADR-010 |
| AD-04 | Rules that change by program and placement type | QR-05, RFP section 7, F-24, CL-004, CL-020, ASM-004, R-001, EU-01, EU-02, QS-07 | Rules are versioned configuration evaluated by small interchangeable evaluators (Strategy). Each decision stores the rule version used. | ADR-004 |
| AD-05 | Authoritative external services that may fail | QR-04, F-17, F-18, ASM-002, ASM-003, ASM-009, R-005, service contracts, QS-05, QS-06 | The core talks to ports, not to services. Adapters translate the contracts, record source timestamps and define the failure behavior of each call. Notifications leave through an outbox so that they never undo a business decision. | ADR-007, ADR-008 |

A trade-off is credible only when it names what gets worse. These are the tensions we resolved and the cost we accept:

| Tension | Resolution | Accepted cost |
|---|---|---|
| Fresh agreement data vs. availability of the directory | Verify the agreement at activation and refuse when it cannot be verified | Activations stop while the directory is down (BR-02). |
| Immediate notification vs. reliable decision | Commit locally, deliver through the outbox | Messages arrive later and may repeat once. |
| Flexible rules vs. simple code | A small vocabulary of rule types configured per type and program | A rule that the vocabulary cannot express needs a new class (ADR-004 revisit). |
| Strong module isolation vs. one transaction | Modules in one process sharing one database, with table ownership | Isolation is enforced by tests, not by the network. |
| Rich history vs. minimum exposure | Audit everything, show history only to the office | External users cannot reconstruct decisions themselves. |
| Complete product vs. one semester | Implement the riskiest slice, design the rest and label it | Four business modules, the closure gate and the web client remain for Evaluation 3 (AR-07). |

# 5. Alternatives and decisions

The same requirements can be built with very different structures. We compared four plausible alternatives against the drivers and the constraints of a three-person team in one semester. This is a qualitative fit analysis, not an invented benchmark.

| Alternative | Main strength | Main weakness for DH-05 | Verdict |
|---|---|---|---|
| Layered monolith | Simple; one transaction protects the gates. | Nothing prevents the 'controller talks to everything' decay across eleven capabilities. | Rejected |
| Modular monolith | Strong internal boundaries, local transactions, one pipeline. | Scales as one unit (acceptable at 600-900 placements per year); boundaries need enforcement. | Chosen (ADR-001) |
| Microservices | Independent deployment and scaling. | The activation gate needs data from several services: distributed transactions for a problem we do not have. | Rejected (ADR-009) |
| Low-code platform | Fast forms, approvals and dashboards. | Relationship and field-level access, versioned rules and custom gates are the core of DH-05 and fit low-code poorly. | Rejected (ADR-009) |

The selected strategy in one sentence: one deployable backend organized in business modules with explicit interfaces, one relational database where every table belongs to one module, ports and adapters at the boundary with the institutional services, in-process domain events with an outbox for side effects, and a worker from the same codebase for delivery. It is the simplest structure that satisfies the important drivers and can evolve when the evidence changes.

Every technology presented in class went through the six questions of the decision filter:

| Proposal | Problem it would solve in DH-05 | Decision |
|---|---|---|
| Use microservices | None identified in DH-05 | Reject |
| Use a service mesh | No service-to-service traffic exists | Reject |
| Use an external message broker (Kafka, RabbitMQ) | Notifications must not block or undo decisions | Reject for now (outbox instead) |
| Use serverless functions for reminders | Deadline reminders are occasional | Reject for now; revisit if reminders grow |
| Add a RAG policy assistant | Students and staff may not understand program rules | Not adopted in this release |
| Use adapters for external services | SIS and Employer Directory are outside our control and fail | Adopt (ADR-007) |
| Process notifications asynchronously | A notification failure must not cancel an activation | Adopt (ADR-008) |
| Configurable rules (Strategy) | Rules change by program and placement type | Adopt (ADR-004) |

## Decision register

| ID | Decision | Drivers | Verified by |
|---|---|---|---|
| ADR-000 | Adopt the Horizon common starter kit as technical baseline | C-01, C-03, C-06, AD-05 | Starter tests still pass (tests/importers, test_health_endpoint.py, test_sample_endpoint.py); the DH-05 files are used by the adapters. |
| ADR-001 | Use a modular monolith as architectural style | AD-01, AD-02, AD-03, AD-05, C-01, C-03 | tests/architecture/test_module_boundaries.py fails when a module imports another one outside the allowed table; one runnable artifact (API + worker). |
| ADR-002 | One relational database with module-owned tables and transactional commands | AD-01, AD-03 | test_failure_in_the_middle_rolls_everything_back; test_two_concurrent_activations_only_one_commits. PostgreSQL run: NOT EXECUTED (TP-14). |
| ADR-003 | Explicit placement lifecycle with gated transitions | AD-01, AD-03 | tests/unit/test_placement_lifecycle.py (6); activation tests (8). Closure gate: DESIGNED only. |
| ADR-004 | Versioned rule configuration evaluated through Strategy | AD-04, QR-05 | test_rule_change_is_configuration_only; test_qs04_audit_reason_records_the_rule_version_used. Eligibility rules: DESIGNED. |
| ADR-005 | Centralized relationship-based access with audience projections | AD-02, QR-01, QR-02 | test_unrelated_user_gets_not_found; test_host_organization_sees_readiness_without_restricted_evidence; test_qs02_*. Evaluation/incident projections: DESIGNED. |
| ADR-006 | Append-only audit history and versioned records | AD-03, QR-03 | test_audit_history_can_only_be_appended_and_read; audit assertions in activation tests; test_one_correlation_id_links_audit_entry_and_notifications. |
| ADR-007 | Ports and adapters for identity, student information and employer directory | AD-05, QR-04, ASM-002, ASM-003 | tests/unit/test_reference_and_adapters.py; test_directory_down_means_activation_refused; architecture test: mocks imported only by M-11 and composition points. |
| ADR-008 | In-process domain events with a transactional outbox for side effects | AD-05, QR-04, AD-03 | tests/integration/test_outbox.py; test_qs06_worker_crash_*; test_qs06_outbox_payload_*. |
| ADR-009 | Do not adopt microservices, service mesh, serverless, low-code or RAG in this release | AD-01 to AD-05, C-01, C-05 | Not a testable decision; reviewed against the decision filter. Revisit triggers recorded in ADR-001. |
| ADR-010 | Concurrent commands: conditional update on status and a correlation id per command | AD-01, AD-03 | test_two_concurrent_activations_only_one_commits; test_one_correlation_id_links_audit_entry_and_notifications; test_refusal_and_denial_carry_machine_readable_codes. Parallel PostgreSQL run: NOT EXECUTED. |

The full records, including ADR-010 (conditional update and correlation id, added after the slice review), are in 'Architecture Decision Records - DH-05'.

### Patterns and principles, with what we do not claim

| Choice | Where | Why it is justified | What it does not do |
|---|---|---|---|
| Strategy + Factory Method | M-06 evaluators and Rule EvaluatorFactory | Rules change by program; a new rule is configuration or one new class | It does not let anyone write arbitrary logic in the JSON file. |
| Adapter (ports and adapters) | M-11 adapters, ports in the shared kernel | Mock contracts and failures stay outside the core | It does not make a real integration correct; contracts must be tested against the real service. |
| Repository + conditional update | SqlPlacementRepository.save | Persistence hidden from the use case; lost update impossible | It does not lock rows; the loser still runs the gate before failing. |
| Unit of work (one session per command) | ActivatePlacement.execute | State, audit and outbox commit together | It does not make the directory call atomic with the write. |
| Observer (in-process events) + outbox | EventDispatcher, on_ placement_activated, deliver_pending | The use case does not know who reacts; delivery failures never undo a decision | It does not prevent one duplicate message after a crash between send and commit. |
| Facade per module | Public functions of each module | Modules do not reach into each other | A folder alone does not enforce it; the architecture test does. |

We reviewed the whole refactoring.guru catalog and the enterprise patterns of the course. State was considered for the lifecycle and Proxy for access control; both were rejected with a written reason (Design Patterns sheet). No problem, no pattern.

# 6. Architecture views

One system, different questions, different views. Each view answers the question of one audience and leaves out what that audience does not need; the same names are used everywhere. All figures share one notation: the traceability ID of each element is on its tab, teal is inside DH-05, plum is people, grey is outside DH-05, dashed means designed but not built, and the saffron highlight marks the critical path of the activation gate (AD-01).

| View | Audience | Question answered |
|---|---|---|
| V-01 Context | Sponsor, office director, IT Services | Who uses DH-05 and which institutional systems surround it? |
| V-02 Container | Architect and developers | Which deployable parts collaborate and where is the data? |
| V-03 Component | Backend team | How is the API divided into modules and which ones |

| View | Audience | Question answered |
|---|---|---|
| | | may call which? |
| V-04 Component (zoom) | Developers of M-03 and M-04 | How does the activation gate work inside the Placements module? |
| V-05 Deployment | Operations / IT Services | Where does the software run and which connections cross trust boundaries? |
| V-DEV Development | Developers | Which package may import which, and what checks it? |
| V-DOM, V-DATA | Developers, reviewers | Which concepts and tables exist, and who owns them? |
| V-TXN, V-STATUS, SEQ-01..05 | Backend team, reviewers | What happens inside a command and when something fails? |
| V-UX-01, V-UX-02 | Product owner, users | How does each role reach its work, and what does the screen show? |

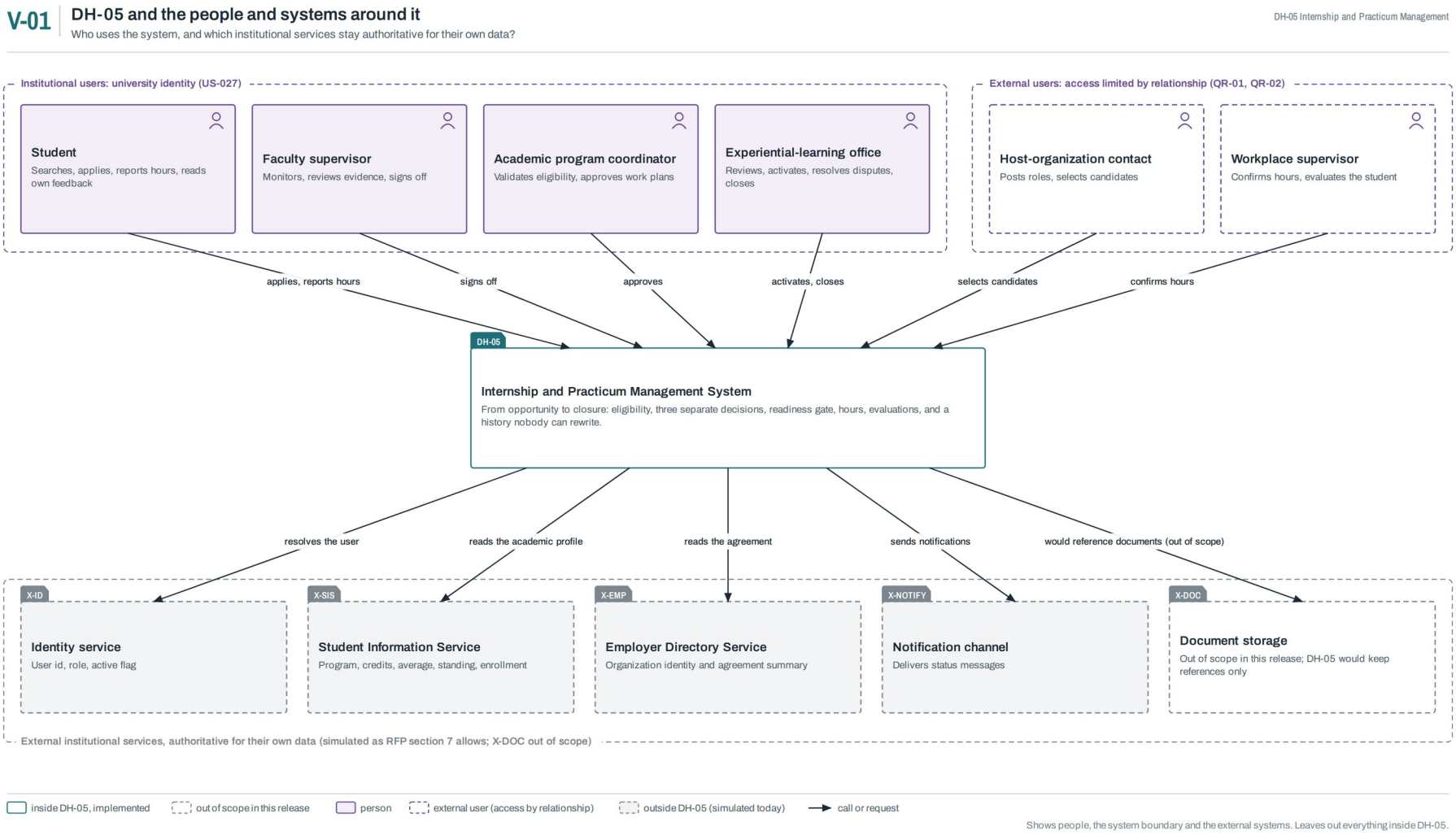



*Figure 1. V-01 Context view.*

The external users have their own boundary because they are the reason for AD-02 and QR-01. Four services are simulated, as the RFP allows, and they stay authoritative: DH-05 reads what it needs for each decision and never keeps the academic record or the agreement (ASM-002, ASM-003). Document storage is out of scope in this release; it is drawn because DH-05 would keep only references to documents.

## Context relationship catalog

| External party | DH-05 expects | DH-05 provides | Failure consequence |
|---|---|---|---|
| Identity (X-ID) | A user id, role and active flag for the caller | The user id in the header (development) or SSO later | 401 if unknown or inactive; 503 if disabled. No local identities are invented. |
| Student Information Service (X-SIS) | Program, credits, average, standing, enrollment, with a timestamp | Read requests only | 503: application kept as pending and retried; 409: manual review; 404: business error (designed for M-02). |
| Employer Directory (X-EMP) | Organization identity and agreement status with valid-until date | Read requests only | Any failure at activation blocks it (fail closed). |
| Notification channel (X-NOTIFY) | Acceptance of a message | Recipient and template; no restricted data | Message stays PENDING and is retried; FAILED after 5 attempts. |
| Document storage (X-DOC) | The documents themselves | Evidence references only | Out of scope in this release; the reference remains valid text. |

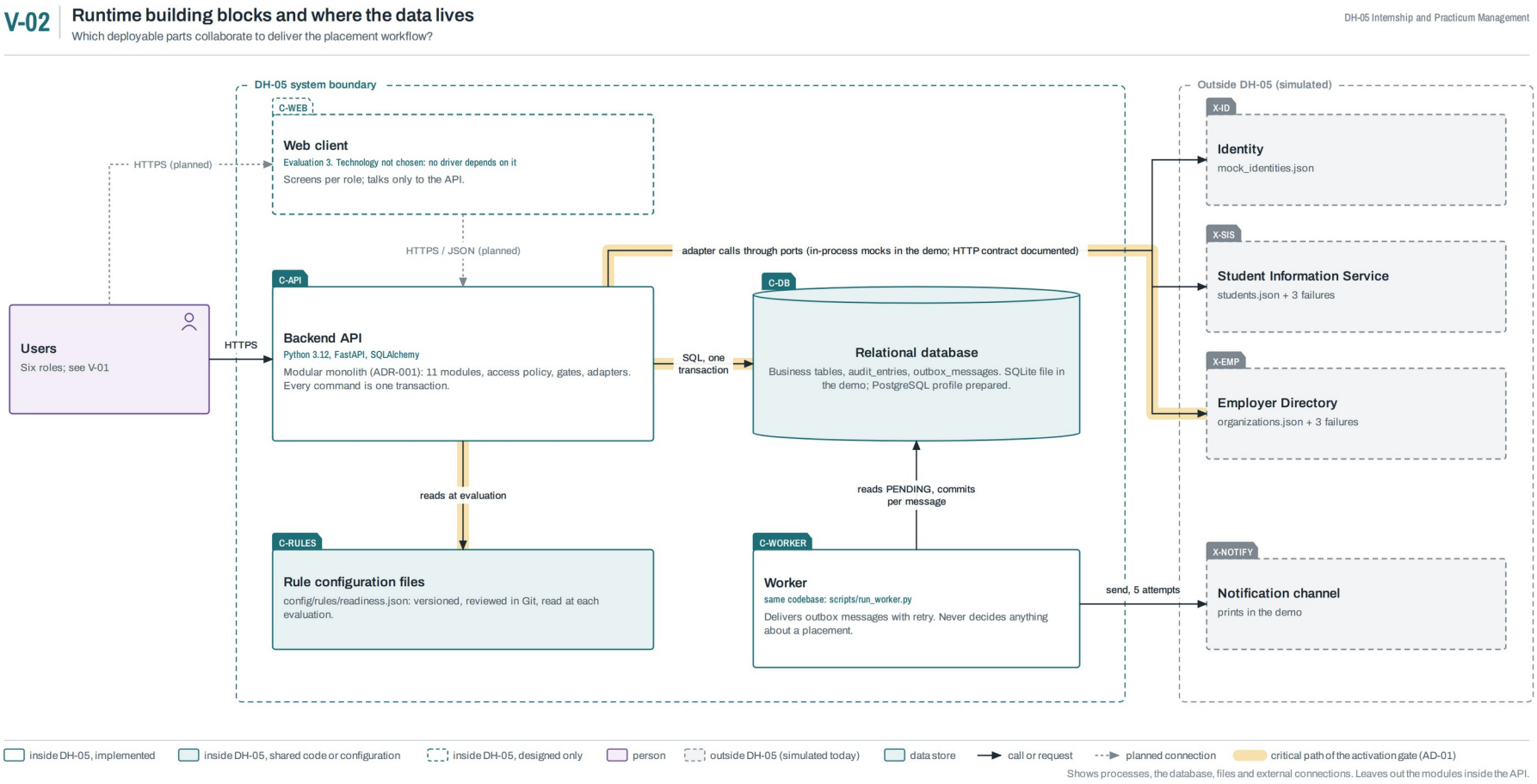



*Figure 2. V-02 Container view.*

The worker is the same artifact started in another mode; it exists only because notifications must never block or undo a decision (ADR-008). The database stores the state change, its audit entry and the outbox message in the same transaction (ADR-002). In the demo the external services are in-process mocks reached through the ports; the HTTP contract is documented for the real services.

## Container responsibilities and non-responsibilities

| Container | Owns | Does not own |
|---|---|---|
| C-API Backend API | Commands and reads, access decisions, gates, rules, adapters, the transaction of each command | Delivering notifications; storing documents; authoritative student or agreement data |
| C-WORKER Worker | Reading PENDING outbox messages, delivery, retries, marking SENT or FAILED | Deciding anything about a placement; re-evaluating the gate |
| C-DB Relational database | Business tables owned by modules, audit_entries, outbox_messages | Business rules; sending messages |
| C-RULES Rule files | The versioned rule configuration | The decisions taken with it (stored by the modules) |
| C-WEB Web client (planned) | Screens per role, input validation for convenience | Authorization decisions; direct database access |

## Modules

A module is a responsibility boundary, not a folder. For every module: the capability it owns, the data it controls, the interface it offers and what it does not own. Status: M-03, M-04, M-06, M-07, M-08, M-09 and M-11 have code; M-01, M-02, M-05 and M-10 are designed.

| Module | Owns | Exposes | Does NOT own |
|---|---|---|---|
| M-01 Opportunities (stages 1-3) | Registration requests, opportunities with their criteria, review results with reason and reviewer. | register_organization, post_ opportunity, review_ opportunity, search_ published(program) | Organization identity and agreement summary (Employer Directory); eligibility rules (M-06). |
| M-02 Applications (stages 4-7) | Applications, eligibility decisions (rule version, per-rule result, source timestamp), employer decisions, academic approvals. | submit_application, request_ self_arranged, record_ employer_decision, record_ academic_approval, application_ status | Student academic data (Student Information Service); readiness conditions (M-04); the placement lifecycle (M-03). |
| M-03 Placements (stages 7, 10-12, 15) | Placements and their status transitions, work-plan versions, supervisor assignments, activation and start dates, closure result. | create_from_application, approve_work_plan, assign_ supervisors, activate, register_ start, close, placement_summary | Readiness evidence (M-04); hours and evaluations (M-05); rules (M-06). |
| M-04 Readiness & Documents (stages 8-9) | Readiness conditions, evidence references, verification results, validity dates. | checklist(placement), evaluate(facts, today) -> Gate Result(rule_version, blocking); register_evidence and verify_ condition are designed | The documents themselves (external document storage, simulated); the agreement master data (Employer Directory). |
| M-05 Monitoring & | Hour entries and reviews, progress reports, incidents with visibility, | report_hours, review_hours, resolve_dispute, report_ | The placement status (M-03); who may read restricted fields |

| Module | Owns | Exposes | Does NOT own |
|---|---|---|---|
| Evaluation<br><br>(stages 13-14) | evaluations (student-visible and restricted fields), signoff. | incident, submit_evaluation, give_signoff, completion_ evidence(placement) | (M-07). |
| M-06 Rule Configuration<br><br>(stages 5, 8, 14) | Rule sets, versions, parameters, reference data (programs with aliases, placement types, document types). | evaluate_eligibility(profile, opportunity), readiness_ requirements(type, program), completion_requirements(type, program), current_version | The decisions themselves (stored by M-02, M-04, M-05). |
| M-07 Access<br><br>(stages All) | Relationships (student of, contact of, supervisor of, coordinator of), access policy, projections. | actor_for(identity), audience(actor, relations), check(actor, action, relations), project_condition(audience, condition) | Identities (Identity service, simulated). |
| M-08 Audit History<br><br>(stages All) | Audit entries. | record(...) inside the current transaction, history(resource); nothing else | Business data. |
| M-09 Notifications & Reminders<br><br>(stages All) | Outbox messages, delivery attempts, reminder schedule. | handles domain events; deliver_ pending() in the worker | Decisions; the notification channel (simulated). |
| M-10 Reporting<br><br>(stages All) | Read-only queries and report definitions. | placements_by(program, organization, status), at_risk() | Any writable data. |
| M-11 Integration Adapters<br><br>(stages All) | Translation of payloads, source timestamps, failure mapping. | Implementations of IdentityPort, StudentInformationPort, EmployerDirectoryPort (the ports are defined in the shared kernel) | Authoritative student, identity or employer data. |

Dependency rules:

1. A module calls another module only through the public classes listed in its 'Exposes' column (today: ReadinessGate, AccessPolicy, AuditRecorder); it never queries another module's tables. Exception by design: M-10 Reporting reads through read-only queries it owns (ADR-002). A single facade object per module is designed, not implemented.
2. The shared kernel (horizon/shared: ports, domain events) may be used by every module. Ports are defined there; only M-11 implements them and only M-11 knows the URLs and payloads of external services.
3. M-06 Rule Configuration, M-07 Access and M-08 Audit History are leaf modules: they import no other module. M-09 and M-11 also import no module (they use the shared kernel).

4. M-03 Placements may use M-04, M-07 and M-08 (implemented). M-04 may use M-06. These rules are checked by tests/architecture/test_module_boundaries.py, which reads the imports of the code.
5. M-03 Placements needs completion evidence from M-05, and M-05 needs to know if a placement is active. To avoid a cycle, M-03 will define CompletionEvidencePort and M-05 will implement it (dependency inversion). Designed, not implemented.
6. Readiness conditions completed by other modules (academic approval, approved work plan) will be updated through domain events, not by direct calls (designed).
7. Access is decided only by M-07, but the audit of a denial is written by the calling use case, because M-07 does not depend on M-08.
8. Routers (api.py) are the composition points: they may wire adapters, mocks and modules together; domain and rule code may not import frameworks or mocks.

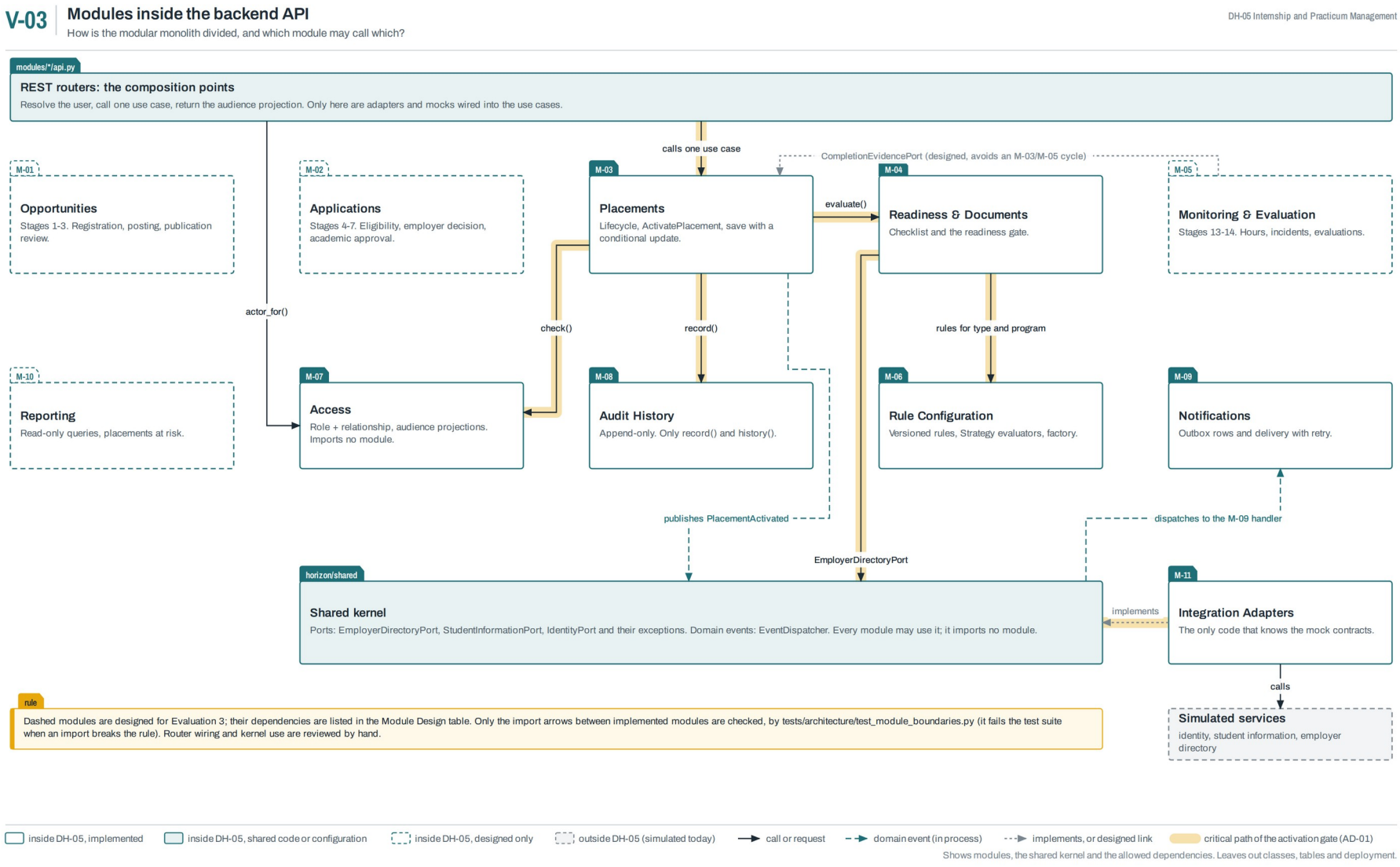



*Figure 3. V-03 Component view of the backend API: modules, shared kernel and allowed dependencies.*

Where does 'accepted hours must reach the required total' belong? In M-05, which owns the hours. M-03 only asks for the completion evidence through CompletionEvidencePort when it evaluates the closure gate; this avoids a cycle between Placements and Monitoring.

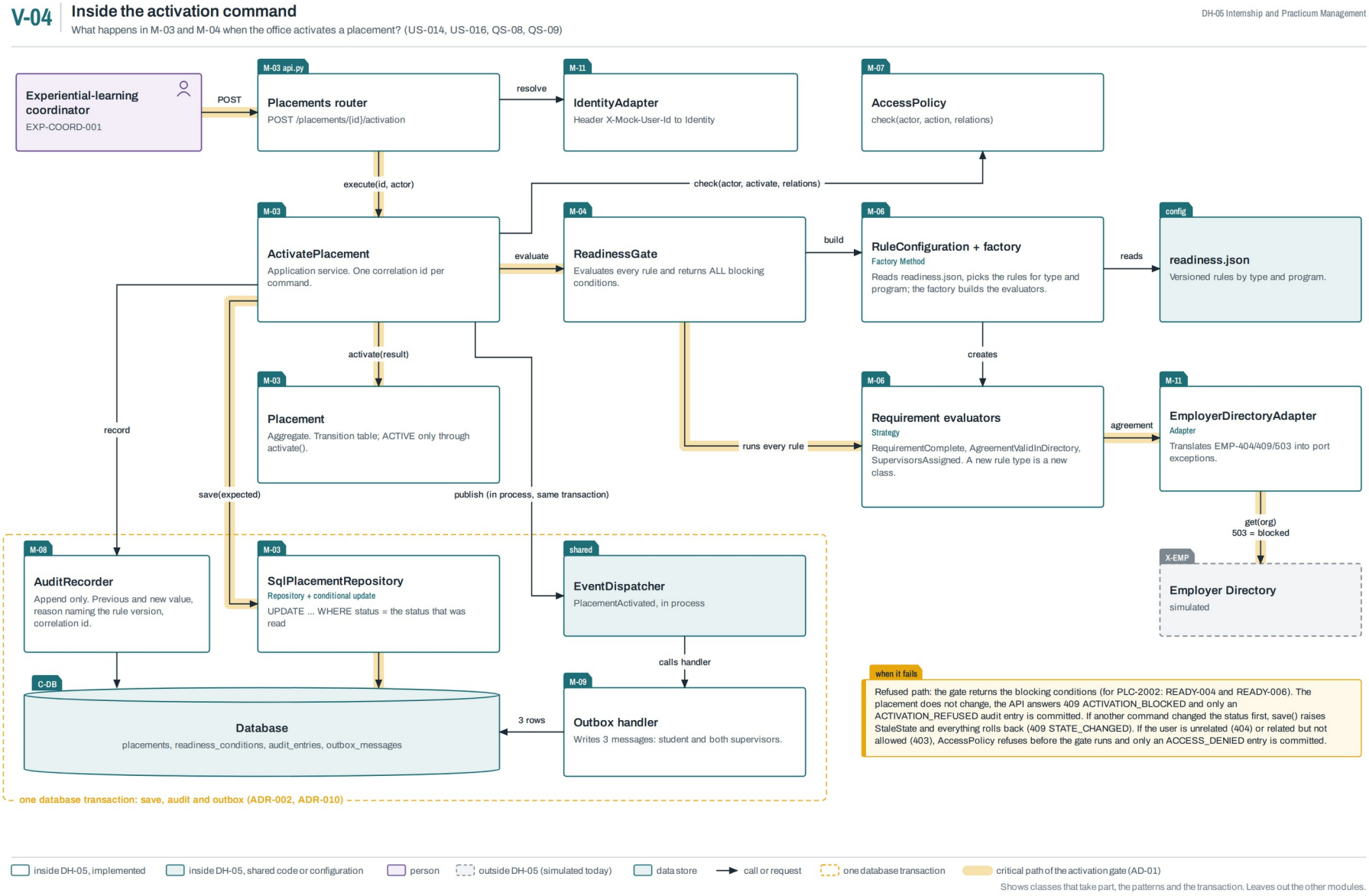



*Figure 4. V-04 Component view (zoom) of the activation command.*

Patterns appear only where a design problem made them necessary: Strategy and Factory Method for rules that change, Adapter at the external boundary, Repository with a conditional update for the write, Observer and outbox for side effects.

## Component specification: ActivatePlacement

| | |
|---|---|
| **Component** | ActivatePlacement (M-03 application service) |
| **Purpose** | Apply the activation command while keeping BR-01, BR-07, BR-11 and BR-12. |
| **Provided operation** | execute(placement_id, actor) -> {placement_id, status, rule_version, correlation_id} |
| **Inputs** | Actor (from IdentityAdapter and AccessPolicy.actor_for), placement id, today and now (injected clocks). |
| **Dependencies** | AccessPolicy (M-07), ReadinessGate (M-04), SqlPlacementRepository and Placement (M-03), AuditRecorder (M-08), EventDispatcher (shared kernel). |
| **State owned** | The placement status and activated_at. |
| **Failure contract** | PlacementNotFound, AccessDenied(reveal), ActivationBlocked(blocking, correlation_id), IllegalTransition, StaleState, any other exception rolls back. |
| **Forbidden** | Calling a mock directly; writing the status without the expected status; committing the audit entry in a separate transaction from the change. |
| **Verification** | TP-04, TP-05; mutation M1 (no conditional update) and M4 (student may activate) are detected. |

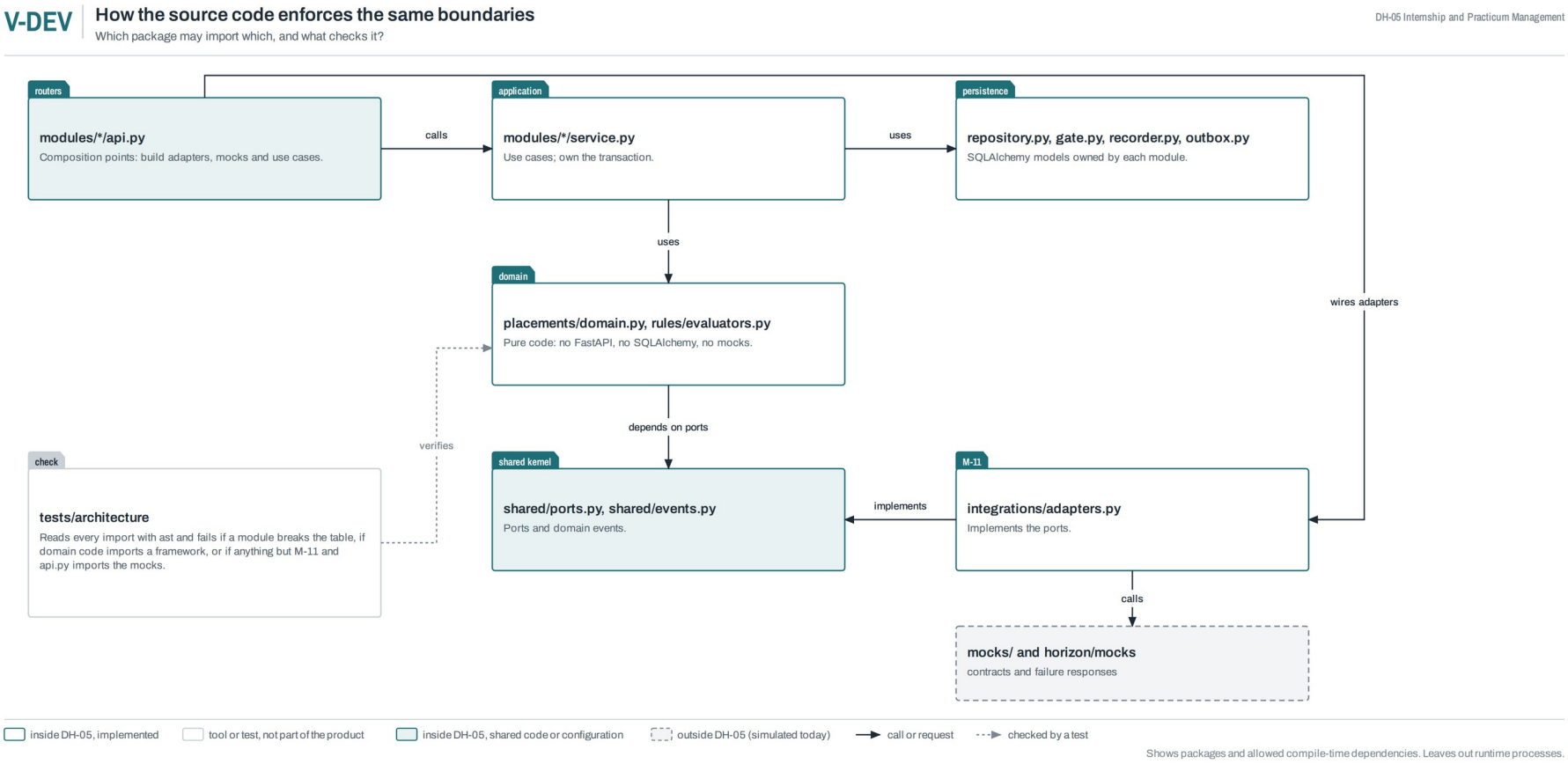



*Figure 5. V-DEV Source-code organization and the test that enforces it.*

Tests detect forbidden imports instead of relying on a README statement: mutation M5 (a forbidden import) makes the architecture test fail.

### Cross-view consistency ledger

| Element | Context | Container | Component / design | Deployment |
|---|---|---|---|---|
| DH-05 | The system | Boundary around C-API, C-WORKER, C-DB, C-RULES | Scope of V-03 and V-04 | Demo laptop or CI runner; pilot host (designed) |
| C-API | Hidden | Backend API | Routers, modules M-01..M-11, shared kernel | uvicorn process |
| C-WORKER | Hidden | Worker | deliver_pending in M-09 | scripts/run_worker.py process |
| C-DB | Hidden | Relational database | Tables owned by M-03, M-04, M-08, M-09 | SQLite file (demo); PostgreSQL (prepared) |
| X-ID, X-SIS, X-EMP | External systems | Outside DH-05 | Ports + M-11 adapters | In-process mocks (demo); real services (pilot) |
| X-NOTIFY | External system | Outside DH-05 | Outbox + worker | Printed by the worker (demo) |

# 7. Domain, data, state and transactions

The model comes from the DH-05 data pack, not from the starter's generic table, and it is divided by module: every table has one owner (ADR-002). We do not copy the export blindly: 'Blocked' is not stored, program codes are normalized and decisions get a rule version.

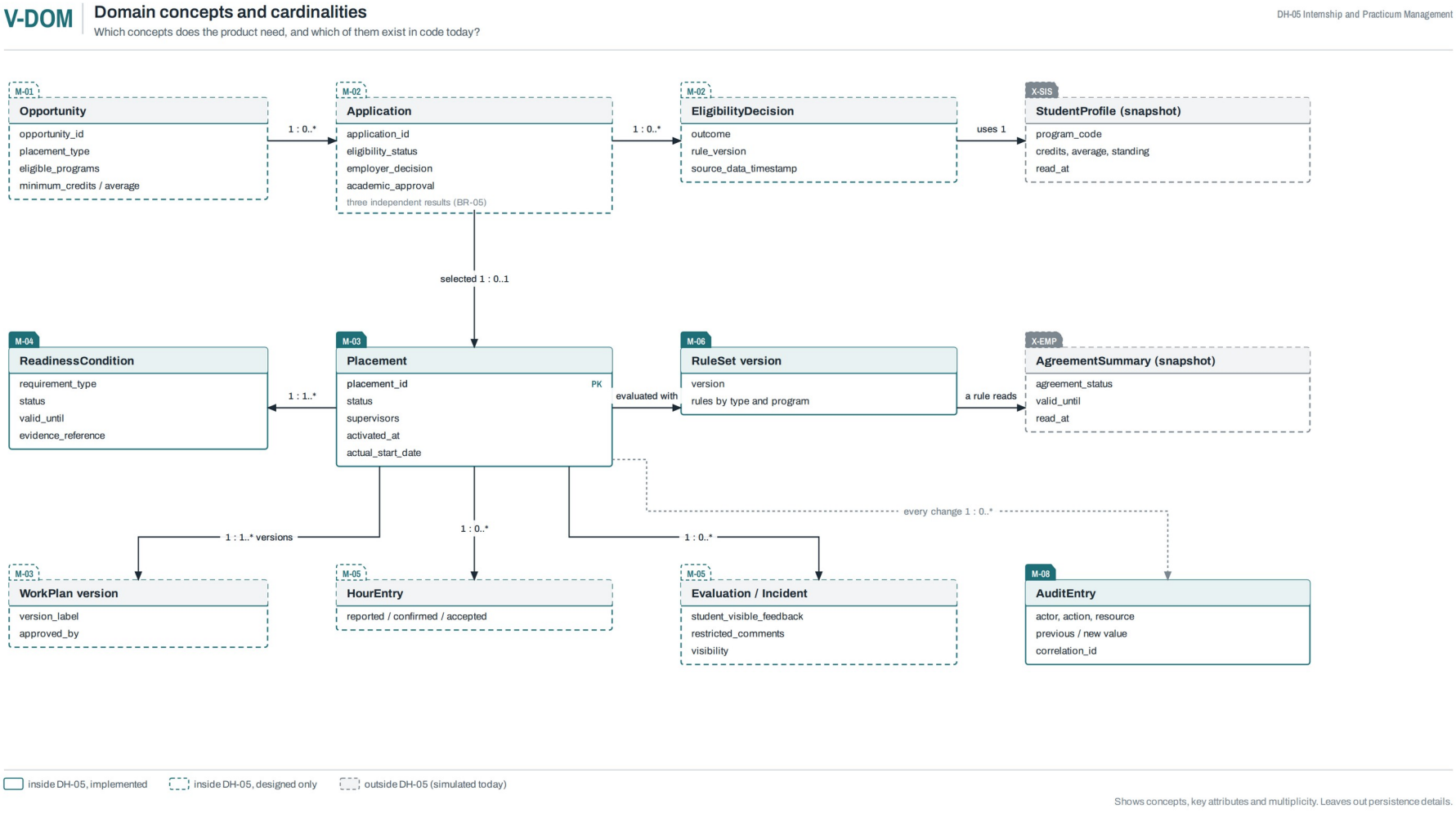



*Figure 6. V-DOM Domain concepts and cardinalities. Dashed concepts are designed for Evaluation 3; grey ones are snapshots owned by external services.*

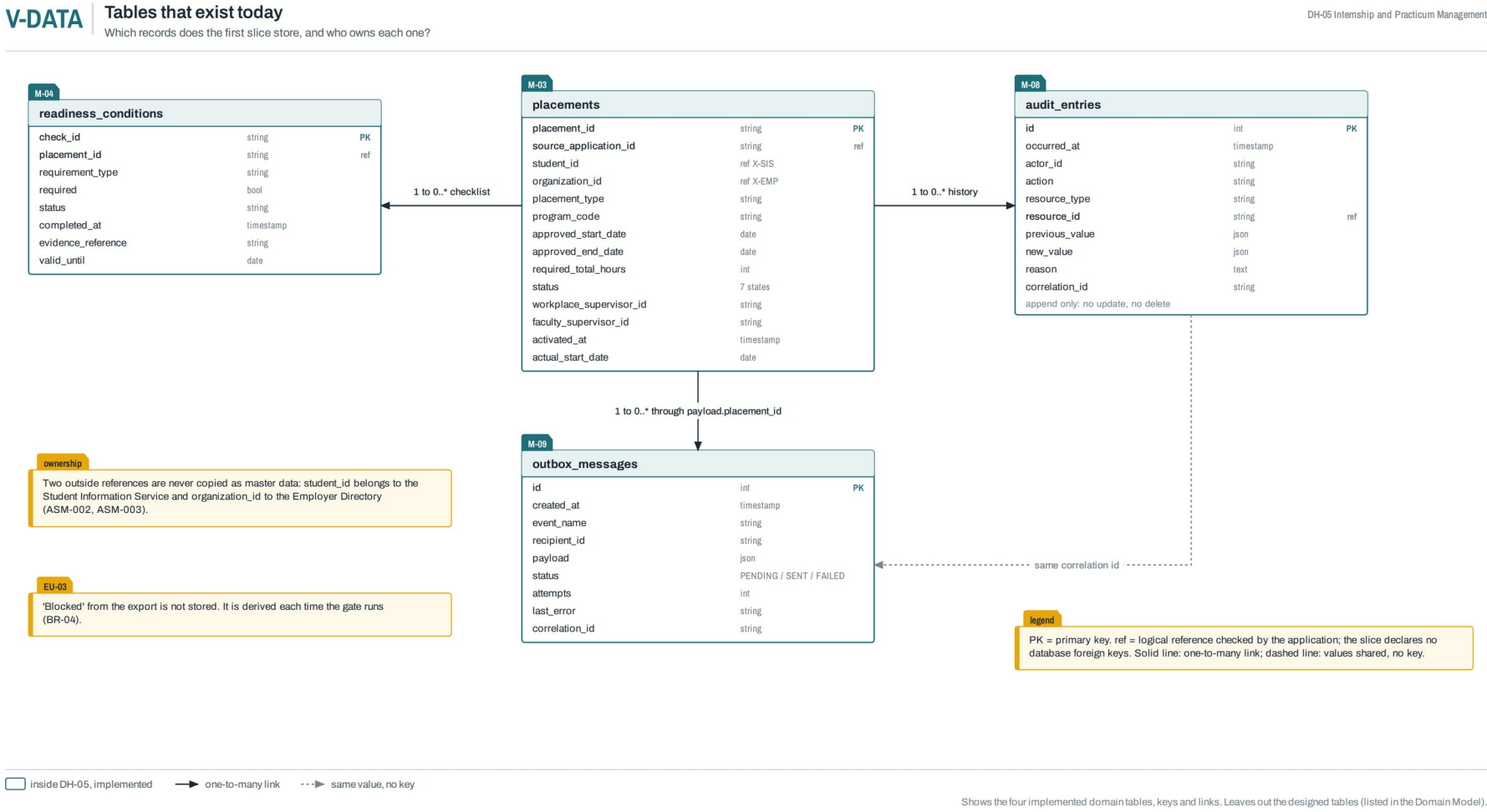



*Figure 7. V-DATA The four domain tables that exist today (the starter's imported_records staging table is not drawn).*

| Entity | Module | Status | Key attributes | Notes |
|---|---|---|---|---|
| Placement | M-03 | IMPLEMENTED | placement_id (PK), source_application_id, student_id, organization_id, placement_type, program_code, approved_start_date, approved_end_date, required_total_hours, status, workplace_supervisor_id, faculty_ supervisor_id, activated_at, actual_start_date | Aggregate root of the lifecycle. Source: placements.csv; 'Blocked' is not stored. |
| Readiness Condition | M-04 | IMPLEMENTED | check_id (PK), placement_id, requirement_type, required, status, completed_at, evidence_ reference, valid_until | Checklist row. Source: readiness_checklist.csv. valid_ until is the expiry used by the gate (null for every seeded row: the export has no such column). |
| AuditEntry | M-08 | IMPLEMENTED | id (PK), occurred_at, actor_id, action, resource_ type, resource_id, previous_value, new_value, reason, correlation_id | Append-only. Written in the same transaction as the change. |
| OutboxMessage | M-09 | IMPLEMENTED | id (PK), created_at, event_name, recipient_id, payload, status, attempts, last_error, correlation_id | PENDING, SENT or FAILED (after 5 attempts). |
| RuleSet (file) | M-06 | IMPLEMENTED | version, readiness rules by placement type and program | config/rules/readiness.json, lookup TYPE@PROGRAM, TYPE, default. |
| Opportunity | M-01 | DESIGNED | opportunity_id, organization_id, title, placement_type, eligible_programs, work_ modality, dates, minimum_credits, minimum_ average, status | Source: internship_ opportunities.csv. |
| Application | M-02 | DESIGNED | application_id, student_id, opportunity_id, application_status, eligibility_status, employer_ decision, academic_approval, current_version | Three separate results (BR-05). Source: applications.csv. |
| EligibilityDecision | M-02 | DESIGNED | review_id, application_id, outcome, criteria_ summary, source_data_timestamp, rule_version | Source: eligibility_reviews.csv; rule_version is new. |
| WorkPlan (version) | M-03 | DESIGNED | work_plan_id, placement_id, version_label, approved_at, approved_by, current_version | New record per version, never overwritten. Source: work_ plans.csv. |
| HourEntry | M-05 | DESIGNED | hour_entry_id, placement_id, work_date, reported_hours, confirmed_hours, accepted_ hours, statuses | Three levels of hours. Source: hour_entries.csv. |
| Evaluation | M-05 | DESIGNED | evaluation_id, placement_id, evaluation_type, student_visible_feedback, restricted_ comments, visibility_classification | Mixed visibility in one record. Source: evaluations.csv. |
| Incident | M-05 | DESIGNED | incident_id, placement_id, category, severity, visibility, status | Visibility drives the projection. Source: placement_incidents. csv. |
| StudentProfile (snapshot) | M-11 / M-02 | IMPLEMENTED (port) | student_id, program_code, enrollment_status, completed_credits, academic_average, | Not stored as master data; owned by the Student |

| Entity | Module | Status | Key attributes | Notes |
|---|---|---|---|---|
| | | | academic_standing, read_at | Information Service. |
| Agreement Summary (snapshot) | M-11 / M-04 | IMPLEMENTED (port) | organization_id, active, agreement_status, agreement_valid_until, read_at | Owned by the Employer Directory. Transient value returned by the port; not persisted. |

## Data dictionary of the implemented tables

| Field | Meaning and rule |
|---|---|
| placements.status | One of the seven lifecycle states. Today only the activate command writes it, through the transition table and conditional update; the other commands are DESIGNED to reuse the same path (ADR-010). |
| placements.activated_at | Instant of the activation commit (UTC). Null before activation. |
| placements.actual_start_date | Real first working day, recorded after activation; never a status. |
| placements.program_code | Normalized through academic_programs.known_aliases (SWE becomes BSC-SWE). |
| readiness_conditions.valid_until | Expiry of the evidence; a COMPLETE row past this date blocks. The exported checklist has no such column, so the seed leaves it null; only synthetic test rows set it. |
| readiness_conditions.evidence_ reference | Reference to a document kept elsewhere; null in projections for RESTRICTED requirements. |
| audit_entries.previous_value / new_ value | JSON snapshots of the changed fields; a refusal stores the blocking list. |
| audit_entries.reason | Free text. Gate decisions name the rule version ('readiness gate readiness-v1' on refusal, 'readiness gate readiness-v1 passed' on success); ACCESS_DENIED entries hold the attempted action (for example placement.activate). |
| audit_entries.correlation_id | Same value as the event, the outbox messages and the API response. |
| outbox_messages.payload | Only placement_id and template; no personal or restricted data. |
| outbox_messages.attempts / status | Incremented on each failed delivery; FAILED at 5. |

## Data ownership

| Information | Authoritative source | What DH-05 keeps |
|---|---|---|
| Student identity and academic profile | Student Information Service | Nothing stored today; the eligibility decision (M-02, designed) will record the values it used and their read_at time |

| Information | Authoritative source | What DH-05 keeps |
|---|---|---|
| Organization identity and agreement summary | Employer Directory Service | Organization reference only; the agreement is read at each gate evaluation and not stored (a refusal records the status that blocked) |
| User identity and institutional role | Identity service | User id in audit entries and relationships |
| Opportunities and reviews | DH-05 (M-01) | Master data |
| Applications, eligibility, employer decision, academic approval | DH-05 (M-02) | Master data; decisions append-only |
| Placement, lifecycle, work-plan versions, supervisors | DH-05 (M-03) | Master data; versions append-only |
| Readiness conditions and evidence references | DH-05 (M-04) | Master data; documents stay in the document system (simulated reference) |
| Hours, progress, incidents, evaluations, signoff | DH-05 (M-05) | Master data; reviews append-only |
| Rule configuration and reference data | DH-05 (M-06) | Versioned configuration |
| Audit history | DH-05 (M-08) | Append-only |

## Invariants and the mechanism that enforces each one

| ID | Invariant | Enforced in | Status | Verified by |
|---|---|---|---|---|
| INV-01 | A placement status is always one of the seven lifecycle states and changes only through the transition table. | Placement._move, TRANSITIONS | VERIFIED | tests/unit/test_placement_lifecycle.py |
| INV-02 | ACTIVE can be reached only by 'activate' with a passed gate result. | Placement.activate | VERIFIED | test_activation_needs_a_passed_gate, test_active_cannot_be_reached_by_other_commands |
| INV-03 | A start date exists only on a placement that was activated. | Placement.register_start | VERIFIED | test_start_date_only_after_activation |
| INV-04 | Each state change is stored only if the stored status still equals the status read (conditional update). | SqlPlacementRepository.save | VERIFIED | test_two_concurrent_activations_only_one_commits |
| INV-05 | State change, audit entry and outbox messages of one command commit together or not at all. | ActivatePlacement.execute | VERIFIED | test_failure_in_the_middle_rolls_everything_back |
| INV-06 | Audit entries can only be appended and read. | AuditRecorder public interface | VERIFIED | test_audit_history_can_only_be_appended_and_read |
| INV-07 | One correlation id links the audit entry, the event and the outbox messages of a command. | new_correlation_id | VERIFIED | test_one_correlation_id_links_audit_entry_and_notifications |

| ID | Invariant | Enforced in | Status | Verified by |
|---|---|---|---|---|
| INV-08 | The seed never modifies the raw data-pack files. | seed_placements | VERIFIED | test_seed_never_modifies_ the_raw_files |

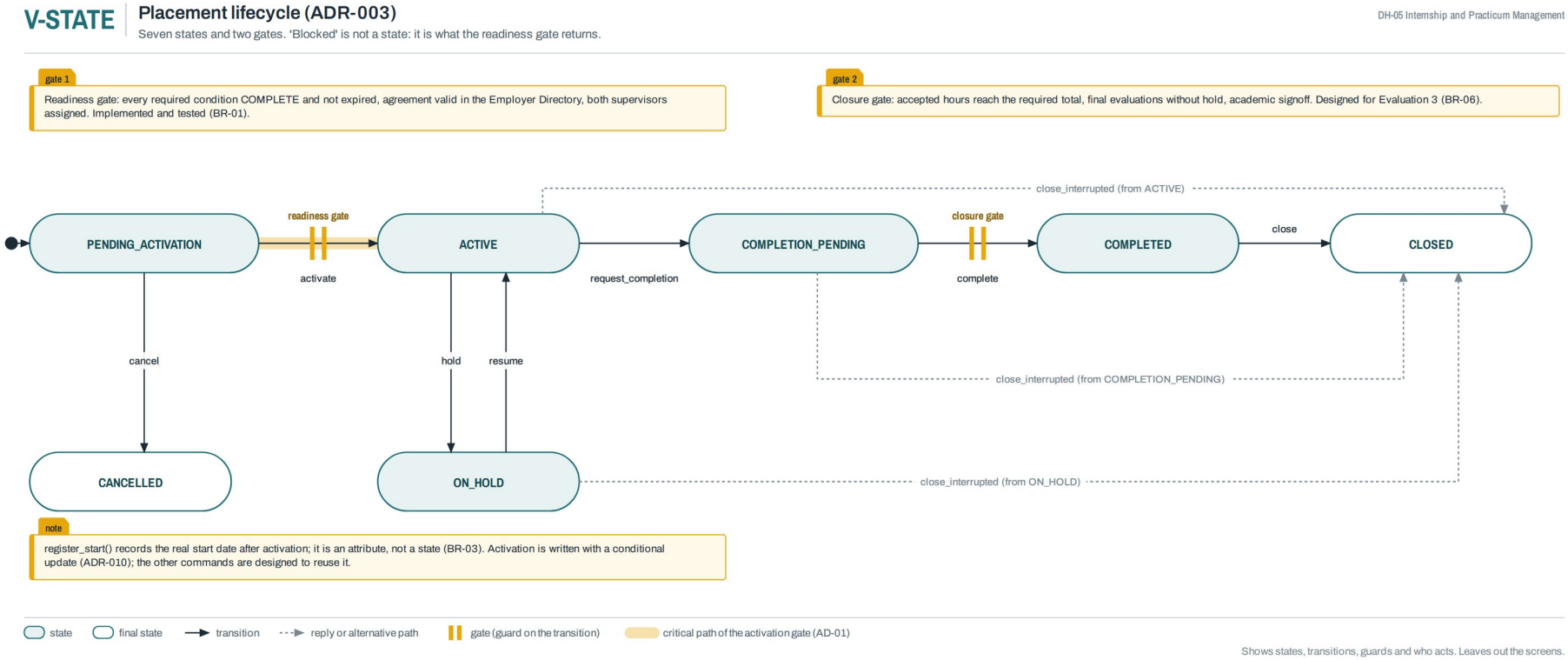



*Figure 8. Placement lifecycle (ADR-003). Two transitions have gates; 'Blocked' is a result of the gate, not a state.*

| From | To | Command | Actor | Guard | Committed side effects | Status |
|---|---|---|---|---|---|---|
| PENDING_ ACTIVATION | ACTIVE | activate | Experiential-learning coordinator | Readiness gate passes; status still PENDING_ ACTIVATION at write | activated_at, audit PLACEMENT_ ACTIVATED, 3 outbox messages | VERIFIED |
| PENDING_ ACTIVATION | (unchanged) | activate refused | Same | Gate blocks | audit ACTIVATION_ REFUSED with blocking list | VERIFIED |
| PENDING_ ACTIVATION | CANCELLED | cancel | Coordinator | Reason required | audit entry | Transition table only |
| ACTIVE | ON_HOLD | hold | Coordinator | Reason required | audit entry | Transition table only |
| ON_HOLD | ACTIVE | resume | Coordinator | Reason required | audit entry | Transition table only |
| ACTIVE | COMPLETION_ PENDING | request_ completion | Student / coordinator | End date reached or final evidence submitted | audit entry, reminders | Transition table only |
| COMPLETION_ PENDING | COMPLETED | complete | Faculty supervisor | Closure gate | audit entry, notifications | DESIGNED |
| COMPLETED | CLOSED | close | Coordinator | Coordinator audit done | audit entry | Transition table only |
| ACTIVE / ON_ HOLD / COMPLETION_ PENDING | CLOSED | close_ interrupted | Coordinator | Reason and result | audit entry | Transition table only |

| Status in placements.csv | Lifecycle state | Note |
|---|---|---|
| Activation pending (PLC-2002) | PENDING_ACTIVATION | Gate shows READY-004 and READY-006 pending |
| Blocked (PLC-2003) | PENDING_ACTIVATION | 'Blocked' is derived from the gate, not stored. For PLC-2003 the gate names READY-007 (expired), READY-008 (missing), the directory agreement EXPIRED and two required conditions not registered |
| ACTIVE (PLC-2001) | ACTIVE | activated_at kept (2027-01-20T08:00:00Z); actual_ start_date not present in the export, left null |
| Completion pending (PLC-2004) | COMPLETION_PENDING | The closure gate (DESIGNED) would show EVAL-002 overdue and EVAL-003 with hold |

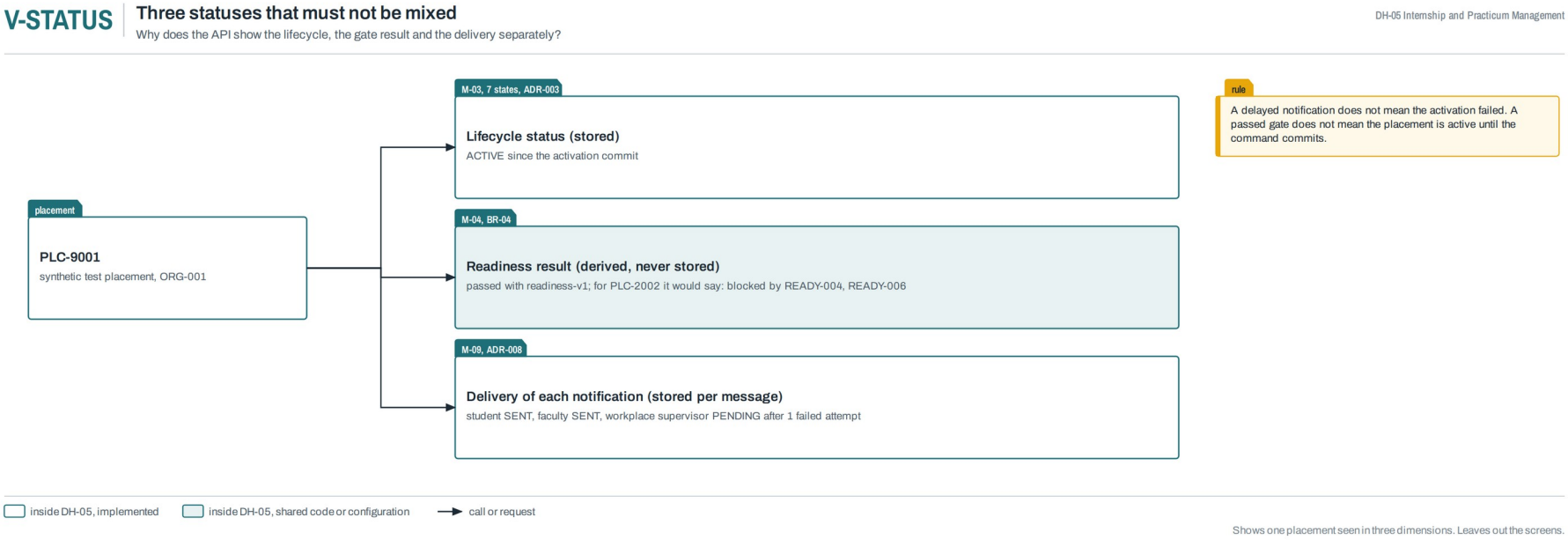



*Figure 9. V-STATUS Lifecycle status, gate result and delivery status are separate facts.*

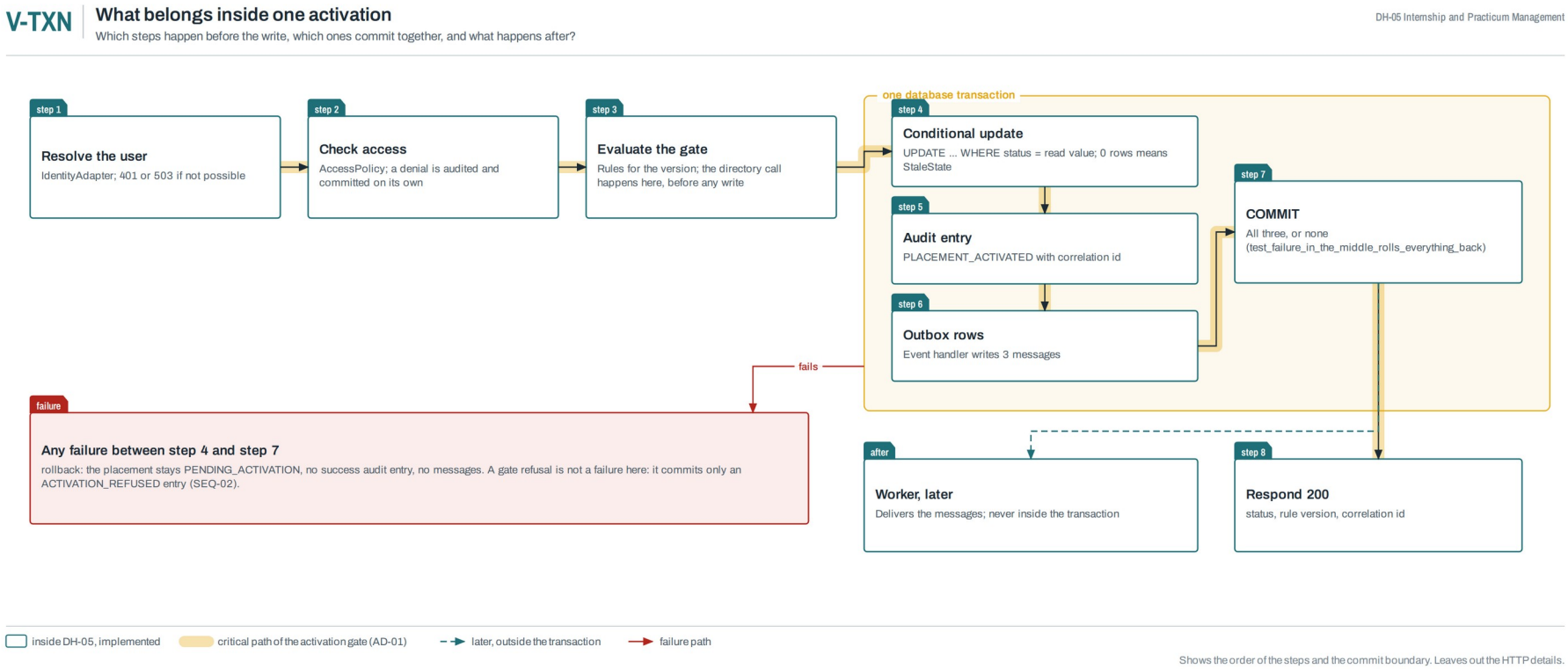



*Figure 10. V-TXN What happens before the write, what commits together, and what happens after.*

The directory call happens before the write phase, never while the write is pending. A refusal by the gate is not a failure of the transaction: it commits only an ACTIVATION_REFUSED entry.

### Data lifecycle

| Data | Rule |
|---|---|
| Placements | Never deleted; CANCELLED and CLOSED remain for program validation. |
| Audit entries | Append and read only through the application; a database administrator could still change rows (AR-05). |
| Outbox messages | SENT and FAILED are kept for diagnosis; an archive policy is not defined yet (BL-12). |
| Rule versions | Old versions stay in Git so every stored version label can be explained. |
| Personal data | Only synthetic data in the course (C-02). Real retention rules must be defined before a pilot. |
| Raw data pack | Never modified (INV-08); every transformation is written to data/generated/transformation_log.md. |
| Schema changes | create_all builds the schema today; versioned migrations are needed before shared data exists (BL-13). |

# 8. Critical interactions and failure behavior

Structure is easy to draw; behavior under failure is where architectures break. Five interactions carry the drivers. Each one was drawn with its failure and each one has a test.

| ID | Interaction | Drivers | Scenario | What it shows |
|---|---|---|---|---|
| SEQ-01 | Activation succeeds | AD-01, AD-03, AD-05 | QS-08 | Access, gate, directory, transition, audit, event and outbox in one transaction. |
| SEQ-02 | Activation refused by the gate | AD-01, AD-03 | QS-08 | Nothing changes except an ACTIVATION_REFUSED audit entry. |
| SEQ-03 | Employer Directory is down | AD-05, AD-01 | QS-08 | Gate stays closed (fail closed). |
| SEQ-04 | Two coordinators activate at the same time | AD-01, AD-03 | QS-09 | Conditional update: one wins, the other gets STATE_CHANGED. |
| SEQ-05 | Notification delivery and failure | AD-05 | QS-06 | Worker delivers after commit; failures are retried, never undo the decision. |

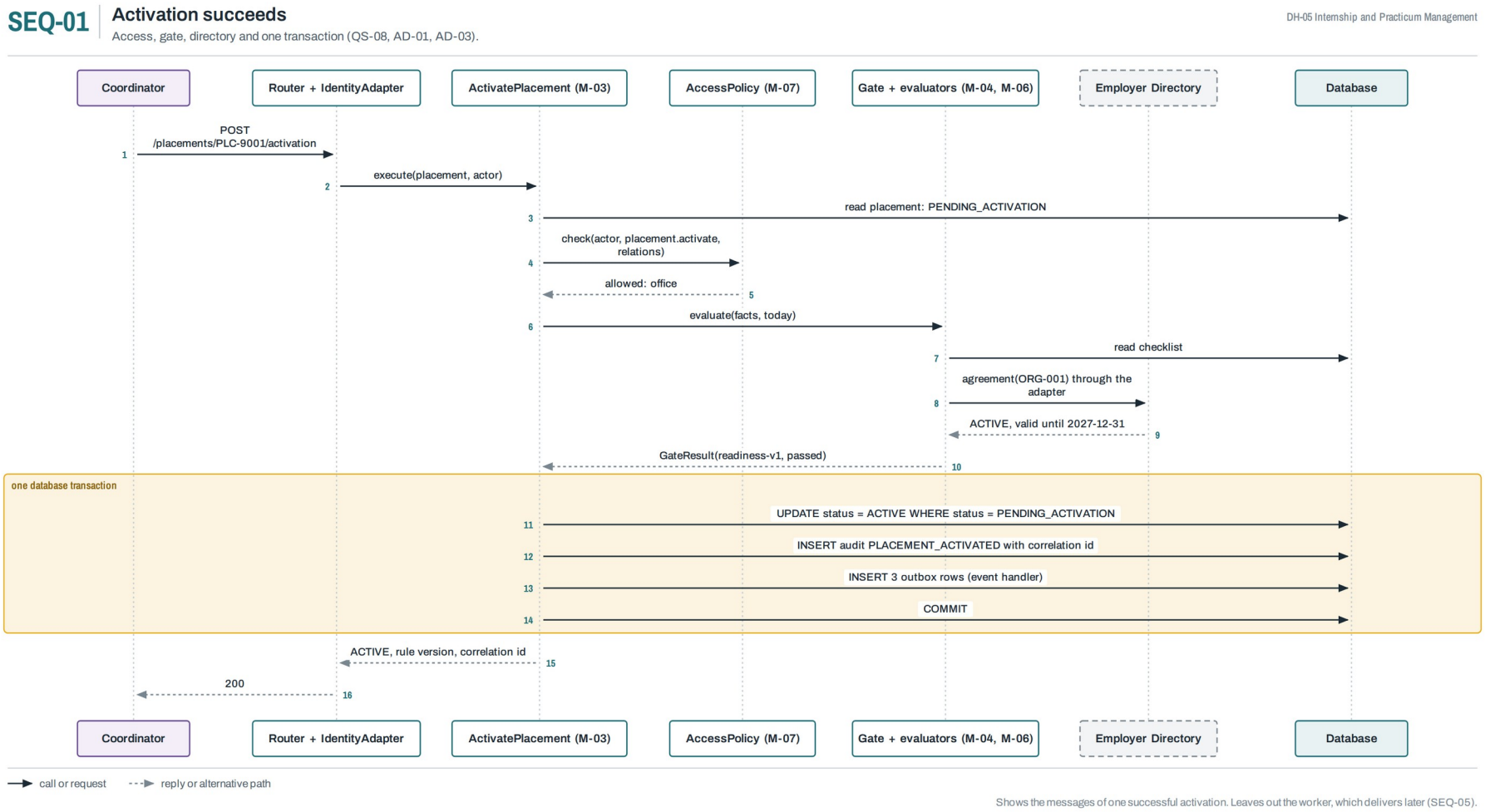



*Figure 11. SEQ-01 Activation succeeds.*

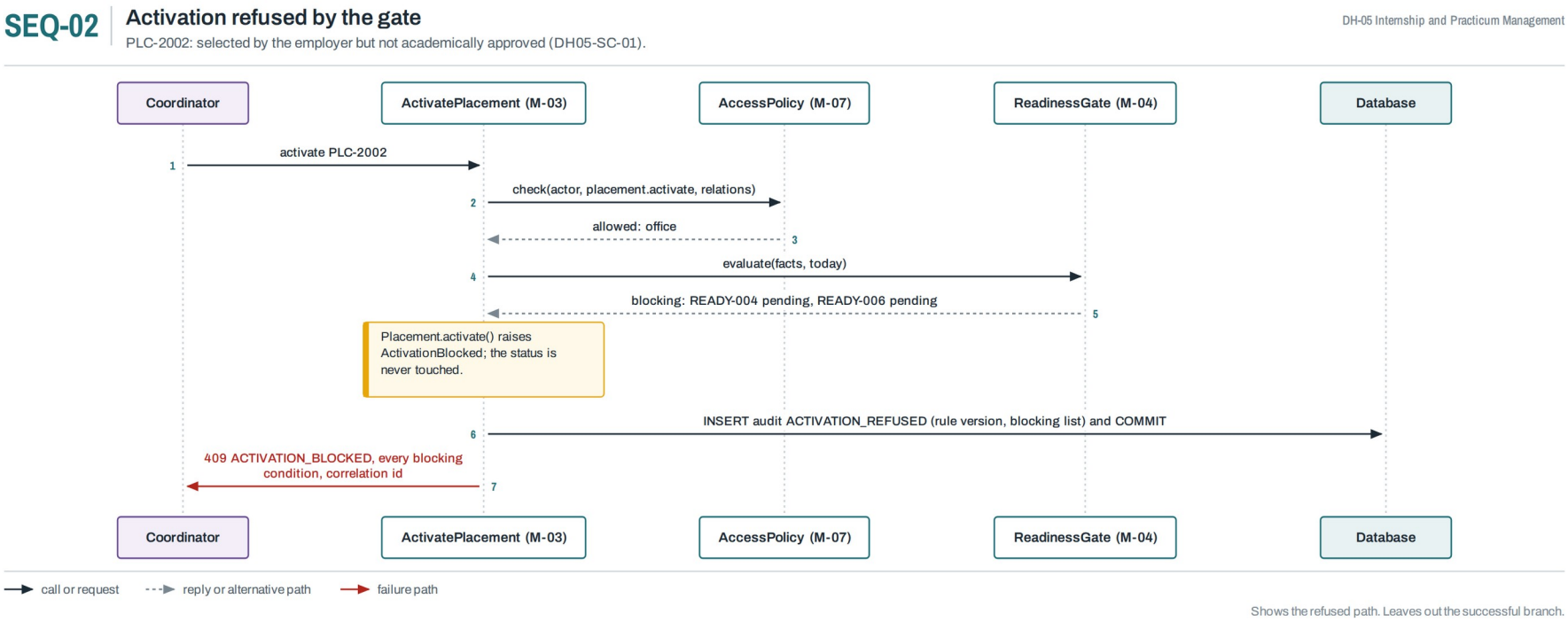



*Figure 12. SEQ-02 Activation refused by the gate.*

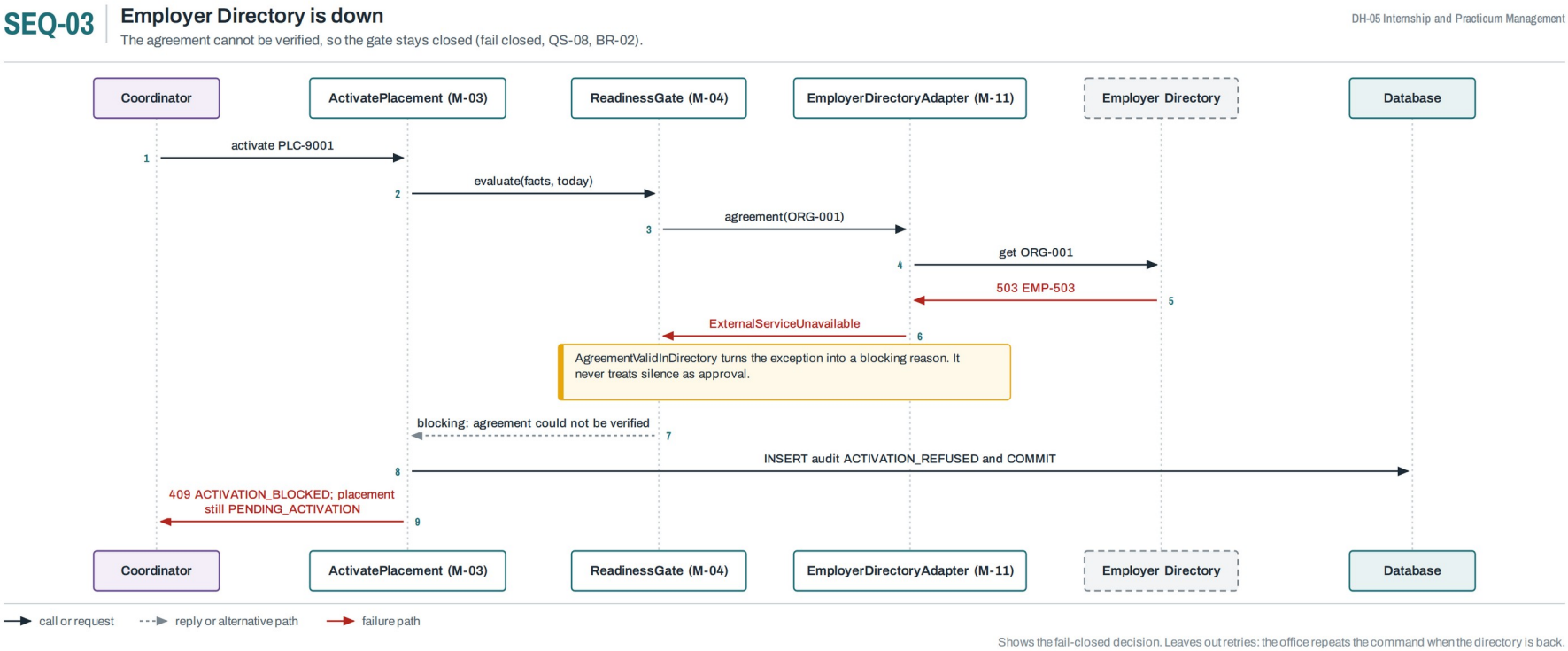



*Figure 13. SEQ-03 Employer Directory is down: fail closed.*

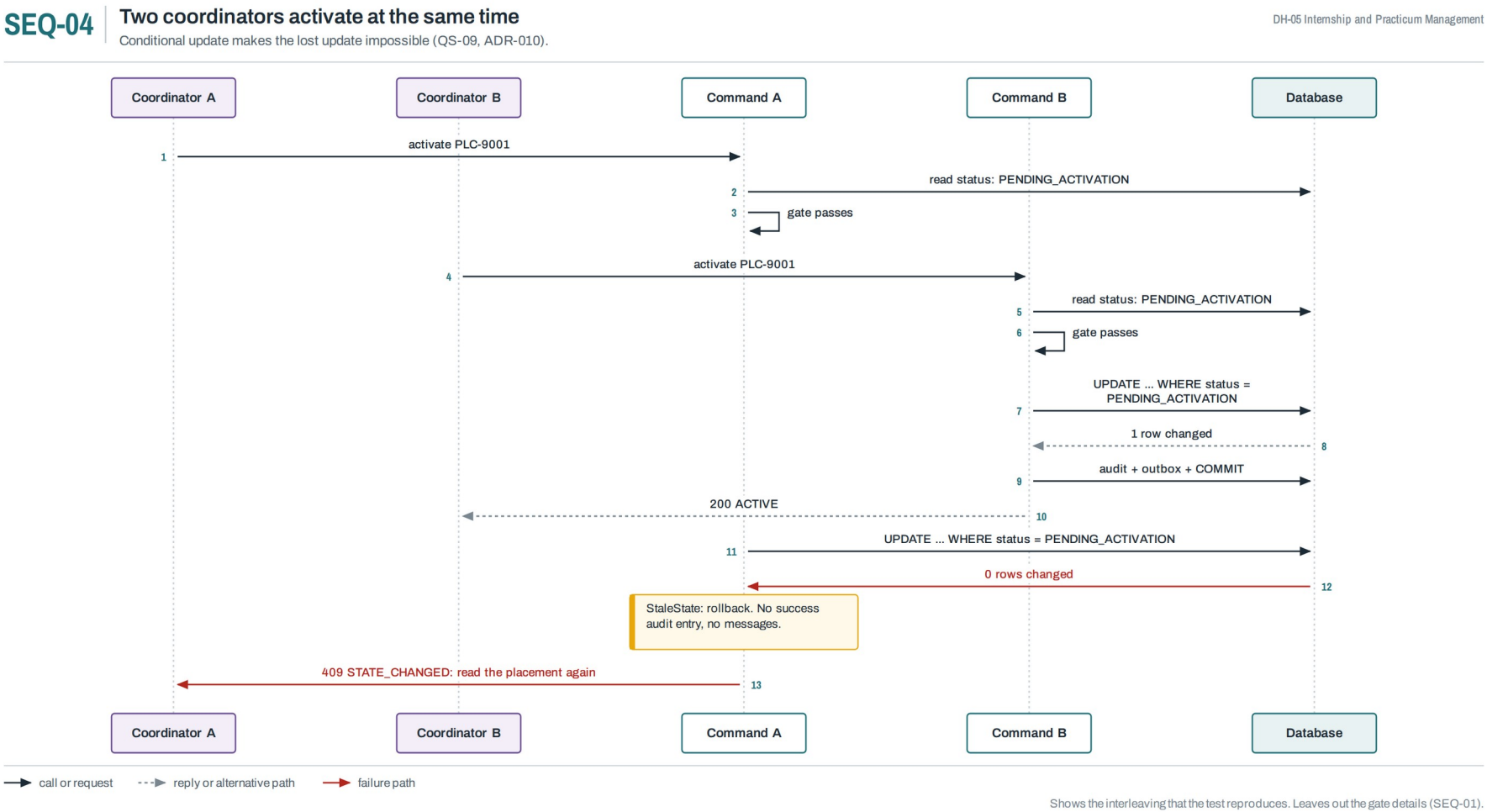



*Figure 14. SEQ-04 Two coordinators activate at the same time (ADR-010).*

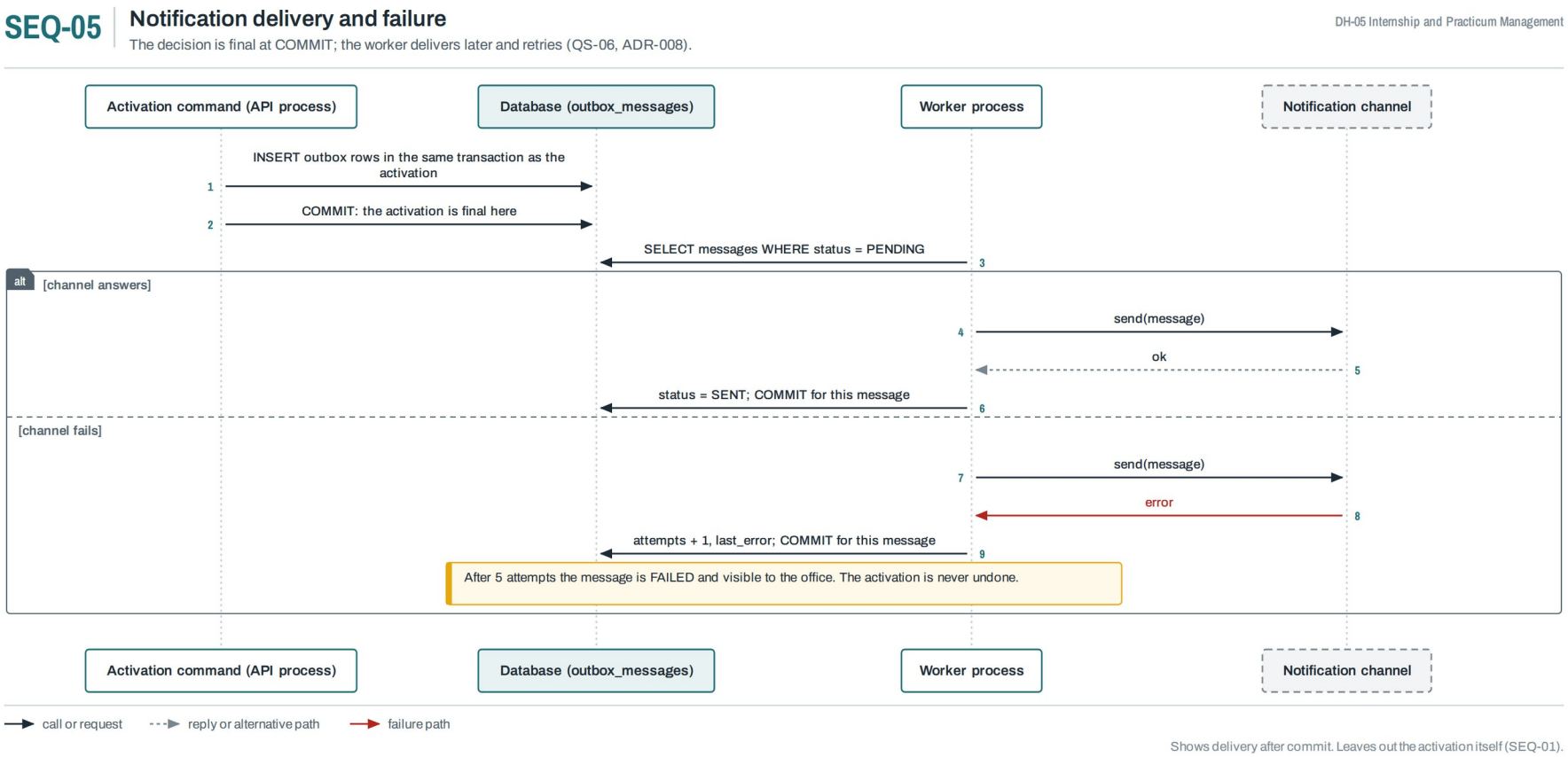



*Figure 15. SEQ-05 Notification delivery and failure.*

The failure matrix makes every claim reviewable: what fails, where it is detected, what the system does, what state is left, who sees what, and which test proves it. Rows marked NOT EXECUTED are honest gaps tracked in the backlog.

| ID | What fails | Detected in | Behavior | State afterwards | Who sees what | Verified by |
|---|---|---|---|---|---|---|
| FM-01 | Required condition PENDING, missing or expired | M-04 evaluator | Refuse: 409 ACTIVATION_ BLOCKED with every blocking condition | Placement unchanged; ACTIVATION_ REFUSED audit entry | Office sees the list and the correlation id | test_sc01_*, test_ expired_condition_ blocks_even_if_ marked_complete |
| FM-02 | Employer Directory 503 / timeout | Agreement ValidIn Directory | Fail closed: refuse, reason 'could not be verified' | Placement unchanged | Office sees the reason and can retry | test_directory_down_ means_activation_ refused |
| FM-03 | Employer Directory 409 (data conflict) | Agreement ValidIn Directory | Refuse, reason 'directory data in conflict, manual review' | Placement unchanged | Office is told to review manually | NOT EXECUTED for the activation path (unit-tested in adapter only) |
| FM-04 | Employer Directory 404 (unknown organization) | Agreement ValidIn Directory | Refuse, reason 'organization unknown to the directory' | Placement unchanged | Office | NOT EXECUTED for the activation path |
| FM-05 | Student Information 503 / 409 / 404 | Student Information Adapter | Translated to port exceptions; M-02 policy: pending + retry / manual review / business error | Application kept | Student sees pending and reason | Adapter translation VERIFIED (3 cases); M-02 policy DESIGNED |
| FM-06 | User without relationship | AccessPolicy | 404 as if the placement did not exist; ACCESS_DENIED audited when the placement exists (an unknown id leaves nothing to audit) | Nothing changes | Caller learns nothing | test_unrelated_user_ gets_not_found, test_ qs02_* |
| FM-07 | User with relationship but not allowed to act | AccessPolicy | 403; audit ACCESS_DENIED | Nothing changes | Caller knows the placement exists | test_student_cannot_ activate_and_denial_ is_audited, test_host_ organization_cannot_ activate_and_denial_ is_audited |
| FM-08 | Another command changed the status first | SqlPlacement Repository. save | 409 STATE_CHANGED; rollback; no success audit, no messages | Winner's state stands | Loser is told to read again | test_two_concurrent_ activations_only_one_ commits (repository), test_api_answers_ state_changed_when_ another_command_ wins (409 through the API) |
| FM-09 | Failure after the state change, before commit | Activate Placement. execute | Rollback of state, audit and outbox | Placement still PENDING_ ACTIVATION | Caller gets an error; retry is safe | test_failure_in_the_ middle_rolls_ everything_back |
| FM-10 | Command repeated after success | Placement. activate | 409 ILLEGAL_TRANSITION; no second audit entry or messages | Unchanged | Caller learns it is already active | test_repeating_the_ command_adds_no_ second_set_of_side_ |

| ID | What fails | Detected in | Behavior | State afterwards | Who sees what | Verified by |
|---|---|---|---|---|---|---|
| | | | | | | effects |
| FM-11 | Notification channel fails | deliver_ pending | Retry up to 5 attempts, then FAILED | Activation committed; message PENDING/FAILED | Failure visible in the outbox | test_failed_delivery_ keeps_activation_and_ retries, test_outbox_ message_is_failed_ after_five_attempts |
| FM-12 | Worker dies in the middle of a run | deliver_ pending | Commit per message: sent messages stay SENT | Remaining messages PENDING | Next run continues | test_qs06_worker_ crash_does_not_ resend_what_was_ already_sent |
| FM-13 | Identity provider disabled or user unknown | get_current_ actor | 503 if disabled; 401 if missing, unknown or inactive | Nothing changes | Caller | test_mock_identity_ disabled_answers_503, test_placements_ endpoint_requires_a_ known_identity |
| FM-14 | Rule configuration file missing or invalid | Rule Configuration | Request fails with a server error; nothing is activated | Unchanged | Operator | NOT EXECUTED (no negative test yet; BL-06) |

# 9. Interfaces and user-facing behavior

Modules talk in two ways: synchronous calls through a facade when the caller needs the answer, and domain events when the publisher does not need to know who reacts.

| Domain event | Published by | Handled by | Reaction |
|---|---|---|---|
| ApplicationSubmitted | M-02 | M-09 | Notify the student and the employer contact |
| EligibilityDecided | M-02 | M-09 | Notify the student with the result and the rule version |
| EmployerDecisionRecorded | M-02 | M-03, M-09 | Create the placement in PENDING_ACTIVATION when the student is selected |
| AcademicApprovalRecorded | M-02 | M-04 | Complete the ACADEMIC_APPROVAL readiness condition |
| WorkPlanApproved | M-03 | M-04, M-09 | Complete the WORK_PLAN readiness condition |
| ReadinessConditionChanged | M-04 | M-09, M-10 | Remind the responsible role; update at-risk report |
| PlacementActivated | M-03 | M-09 | Notify student and both supervisors |
| PlacementStarted | M-03 | M-09 | Start hour and evaluation reminders |
| HoursReviewed | M-05 | M-09 | Notify the student of a confirmation or dispute |
| EvaluationOverdue | M-09 (scheduler) | M-09, M-10 | Reminder to the evaluator; at-risk report |
| PlacementClosed | M-03 | M-09, M-10 | Notify participants; update reports |

Endpoint catalog. Only the two activation-slice operations are implemented; the rest is the proposed contract for Evaluation 3.

| API operation | Module | Allowed caller | Stories |
|---|---|---|---|
| POST /opportunities | M-01 | Host organization contact | US-002 |
| POST /opportunities/{id}/review | M-01 | Experiential-learning coordinator | US-003 |
| GET /opportunities?program= | M-01 | Student | US-004 |
| POST /applications | M-02 | Student | US-005, US-007 |
| POST /applications/{id}/employer-decision | M-02 | Host organization contact | US-009 |
| POST /applications/{id}/academic-approval | M-02 | Academic program coordinator | US-011 |
| POST /placements/{id}/work-plans | M-03 | Academic program coordinator | US-010, US-021 |
| GET /placements/{id}/readiness | M-04 | Related users only: office, faculty supervisor, workplace supervisor, student, host organization (projection) | US-012 |
| POST /placements/{id}/activation | M-03 | Experiential-learning coordinator | US-014, US-016 |
| POST /placements/{id}/hours | M-05 | Student | US-017 |
| POST /hours/{id}/review | M-05 | Workplace supervisor | US-018 |
| POST /placements/{id}/closure | M-03 | Experiential-learning coordinator | US-025 |
| GET /placements/{id}/history | M-08 | Office, auditor | US-026 |

## Contract: POST /placements/{placement_id}/activation (IMPLEMENTED)

| | |
|---|---|
| **Stories** | US-014, US-016 |
| **Allowed caller** | Experiential-learning coordinator (staff role EXPERIENTIAL_LEARNING_COORDINATOR) |
| **Request** | Header X-Mock-User-Id (required in development). No body. |
| **Idempotency** | Not idempotent by design: a second call after success returns 409 ILLEGAL_TRANSITION and adds no side effects (FM-10). |
| **Versioning** | No version prefix yet; breaking changes require a new ADR (BL-08). |

| Status | Body | Meaning |
|---|---|---|
| 200 | {placement_id, status: 'ACTIVE', rule_version, correlation_ id} | Gate passed; state, audit and outbox committed. |

| Status | Body | Meaning |
|---|---|---|
| 401 | {detail} | Header missing, user unknown or inactive. |
| 403 | {detail} | User related to the placement but not allowed to activate. Audited. |
| 404 | {detail} | Placement does not exist OR user has no relationship (indistinguishable). Audited only when the placement exists. |
| 409 | {code: ACTIVATION_BLOCKED, detail, blocking: [{requirement, check_id, reason}], correlation_id} | At least one rule blocks. Audited as ACTIVATION_ REFUSED. |
| 409 | {code: STATE_CHANGED, detail} | Another command changed the placement first. Read it again. |
| 409 | {code: ILLEGAL_TRANSITION, detail} | Placement is not in PENDING_ACTIVATION. |
| 503 | {detail} | Identity provider disabled. |

## Contract: GET /placements/{placement_id}/readiness (IMPLEMENTED)

| | |
|---|---|
| **Stories** | US-012 |
| **Allowed caller** | Office, faculty supervisor, workplace supervisor, student, host organization (related users only) |
| **Request** | Header X-Mock-User-Id. |
| **Idempotency** | Safe read; the gate is evaluated at read time, so the result can change between calls. |
| **Versioning** | As above. |

| Status | Body | Meaning |
|---|---|---|
| 200 | {placement_id, status, audience, rule_version, ready, blocking[], conditions[{check_id, requirement_type, status, evidence_reference}]} | evidence_reference is null for RESTRICTED requirements outside the office. |
| 401 | {detail} | As above. |
| 404 | {detail} | Not found or no relationship. Denial audited when the placement exists. |
| 503 | {detail} | Identity provider disabled. |

## Errors tell the caller how to recover

| Status | Code | Meaning | Recovery |
|---|---|---|---|
| 401 | - | Header missing, user unknown or inactive | Sign in again; do not retry automatically. |
| 403 | - | Related to the placement but not allowed to act | Do not retry; ask the office. |
| 404 | - | Placement missing or no relationship (indistinguishable on purpose) | Check the identifier; no retry. |
| 409 | ACTIVATION_BLOCKED | At least one rule blocks; the list is in the body | Complete the named conditions, then activate again. |
| 409 | STATE_CHANGED | Another command changed the placement first | Read the placement again; do not resend blindly. |
| 409 | ILLEGAL_TRANSITION | The placement is not in PENDING_ ACTIVATION | Read the current status; nothing to do if already ACTIVE. |
| 503 | - | Identity provider disabled | Retry after the identity service is configured. |

## Event, outbox and port contracts

The event as it is published today (in process):

```
{
  "name": "PlacementActivated",
  "occurred_at": "2026-09-24T14:05:00Z",
  "payload": {
    "placement_id": "PLC-9001",
    "student_id": "HU-STU-5002",
    "workplace_supervisor_id": "SUP-EXT-002",
    "faculty_supervisor_id": "FAC-SUP-101",
    "correlation_id": "<uuid hex, one per command>"
  }
}
```

The current event has no schema version and no event id. It never leaves the process, so this is acceptable today; before a module is extracted or an external consumer appears, add eventId and schemaVersion (BL-11).

| Contract | Owner | Content | Guarantees and limits |
|---|---|---|---|
| Placement Activated | M-03 | payload: placement_id, student_id, workplace_supervisor_id, faculty_ supervisor_id, correlation_id | Handlers run inside the command's transaction and may only write local rows. Order of handlers is not guaranteed. Delivery is in-process and at-most-once: a crash before commit means the event never happened. |

| Contract | Owner | Content | Guarantees and limits |
|---|---|---|---|
| Outbox message | M-09 | event_name, recipient_id, payload {placement_id, template}, status PENDING/ SENT/FAILED, attempts, correlation_id | At-least-once towards the channel: a crash between send and commit may repeat one message; the receiver must tolerate a duplicate (documented limitation). |

| Port | Failure contract |
|---|---|
| EmployerDirectoryPort.agreement(organization_id) -> AgreementSummary | Raises ExternalServiceUnavailable (503), ExternalDataConflict (409), ExternalRecordNotFound (404). Implemented by Employer DirectoryAdapter (M-11). |
| StudentInformationPort.profile(student_id) -> Student Profile | Same three exceptions. Carries read_at. Implemented by Student InformationAdapter (M-11). |
| IdentityPort.resolve(user_id) -> Identity \| None | None for unknown users. Implemented by IdentityAdapter reading mock_identities.json. |

## Integrations

For each external system we answered the eight questions of the course integration checklist (full answers in the Integration Design sheet). Failure behavior is a business decision: fail soft when a request can wait, fail closed when going ahead without the answer would put the student and the university at risk (for example, activating without a valid agreement).

| System | Authoritative for | Communication | Failure behavior |
|---|---|---|---|
| Identity service (simulated) | Identity service owns user id, display name, e-mail, institutional role and active flag. | Header X-Mock-User-Id resolved by the Identity Adapter (reads mock_ identities.json). Real SSO later. | Fail closed: no identity, no access. |
| Student Information Service (simulated) | Authoritative for program, period, enrollment, credits, average, standing and previous placements (ASM-002). | In-process call to the mock service class through Student InformationAdapter (same payload as the contract GET / mock/student-information/ students/{student_id}); HTTP when the real service exists. | 404 -> business error; 409 -> manual review; 503 -> pending + retry (fail soft). |
| Employer Directory Service (simulated) | Authoritative for organization identity, active status, sector, country and agreement summary (ASM-003). | In-process call to the mock service class through EmployerDirectoryAdapter (same payload as the contract GET / mock/employers/{organizatio n_id}); HTTP when the real | 404 -> activation refused, 'organization unknown to the directory'; 409 -> activation refused, 'directory data in conflict, manual review' (FM-03); 503 -> activation refused, |

| System | Authoritative for | Communication | Failure behavior |
|---|---|---|---|
| | | service exists. | 'could not be verified' (FM-02). |
| Notification channel (simulated) | DH-05 owns the message content; the channel only delivers. | Outbox table read by the worker process. | Retry up to 5 attempts, then FAILED and visible in the outbox; an alert to the office is DESIGNED. |

## User-facing behavior (designed)

| Interaction | Design rule |
|---|---|
| Readiness panel | Show every blocking condition at once with its id and the role that must act; never only the first one. |
| Activate button | Disabled with the reason while the gate blocks; the API still decides (BR-01). |
| Conflict | On 409 STATE_CHANGED, reload the placement and explain that someone else acted; keep no stale data on screen. |
| Restricted evidence | External audiences see 'Complete (reference hidden)', never an empty field that looks like a bug. |
| Status words | Lifecycle, gate result and delivery are three labels, written in words, not color alone. |
| Double click | The UI may prevent it, but correctness comes from conditional update and ILLEGAL_ TRANSITION. |

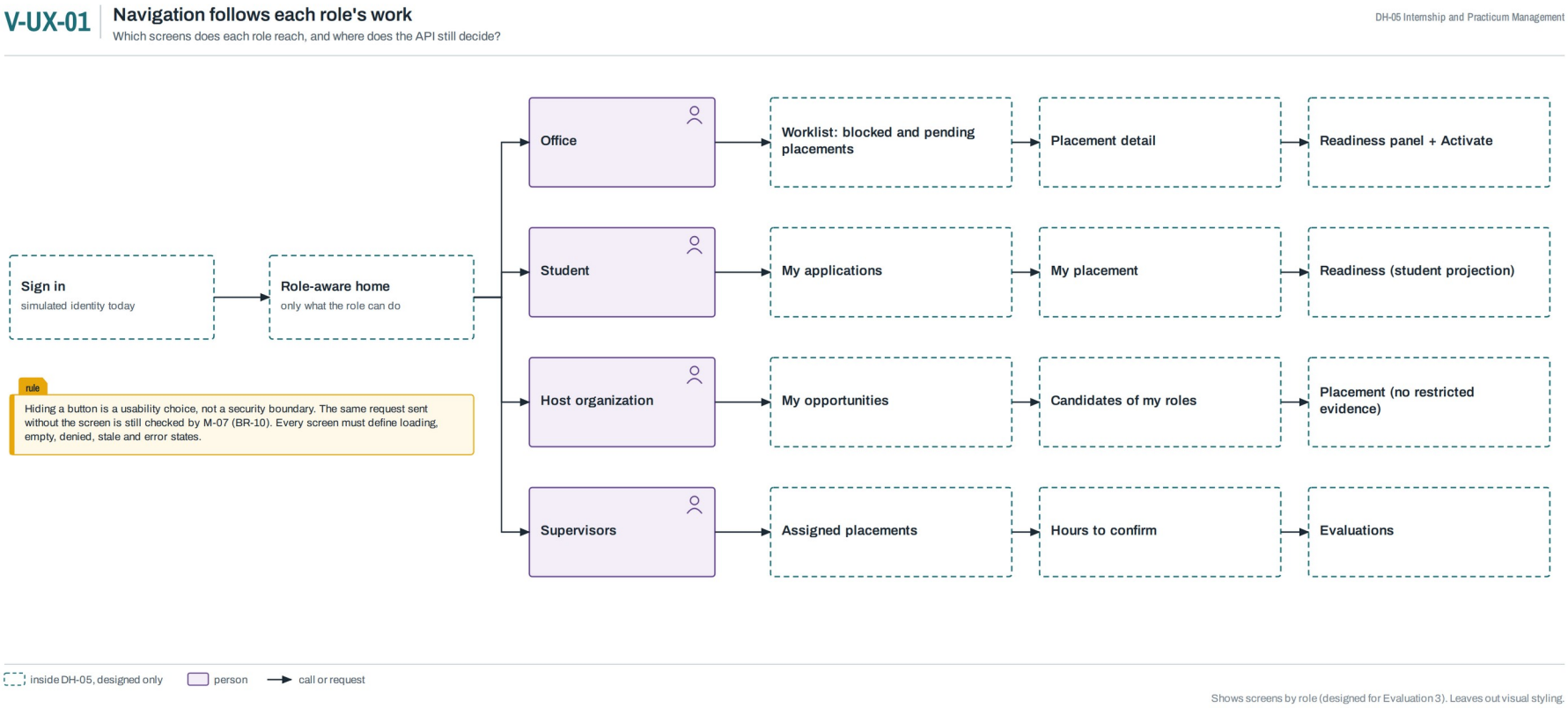



*Figure 16. V-UX-01 Navigation by role.*

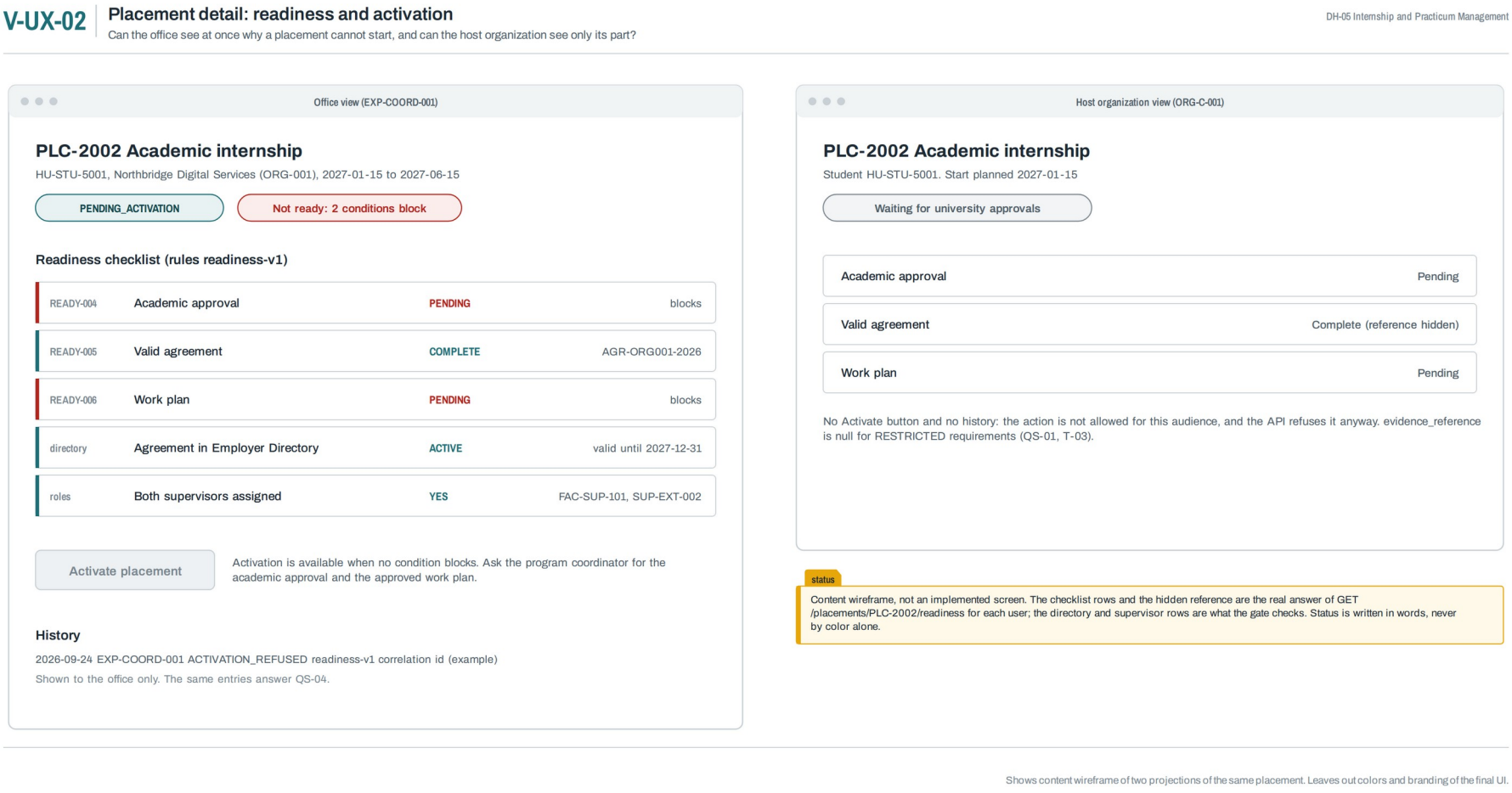



*Figure 17. V-UX-02 Content wireframe of the placement detail for two audiences.*

The office sees every blocking condition and why Activate is disabled; the host organization sees only its projection. The usability check for this screen is planned, not run.

# 10. Security and privacy

Access is decided in one place (M-07) with role plus relationship, and every response goes through an audience projection. Authentication only establishes who the caller is; it never authorizes an action by itself. The matrix is deliberately conservative: where CL-018 is open, access is denied until the Data Privacy Officer answers.

| Resource | Student | Host org. | Workplace sup. | Faculty sup. | Program coord. (DESIGNED) | Office |
|---|---|---|---|---|---|---|
| Published opportunity | Read if their program is eligible | Read/edit own organization's | - | Read | Read | Read/review |
| Application (status, documents APPLICATION) | Own applications | Candidates of own opportunities | - | - | Own program | All |
| Student academic profile (credits, average, standing) | Own | Never | Never | Own placements | Own program | All |
| Placement summary and readiness checklist | Own | Own organization, without RESTRICTED evidence | Assigned placements | Assigned placements | Own program | All |
| RESTRICTED documents (agreement, insurance, confidentiality) | Status only | Status only | - | Status only | Status only | Full |
| Hour entries | Own (all levels) | - | Assigned placements | Assigned placements | Own program | All |
| Evaluation - student-visible feedback and rating | Own | Own evaluations submitted | Own evaluations | Assigned placements | Own program | All |
| Evaluation - restricted comments | Never | Never (CL-018 open) | Own comments only | Assigned placements | Pending CL-018 (deny) | All |
| Incident with visibility STUDENT/PLACEMENT_TEAM | If reporter or placement member | - | Assigned placements | Assigned placements | Own program | All |
| Incident with visibility RESTRICTED or INTERNAL | Only if they reported it | Never | Never (CL-018 open) | Assigned placements | Pending CL-018 (deny) | All |
| Audit history | - | - | - | - | Own program (read) | All (read) |

Implemented today (activation slice): 'office' means the staff roles EXPERIENTIAL_LEARNING_COORDINATOR, PROGRAM_DIRECTOR and AGREEMENTS_REPRESENTATIVE, and only the first may activate. The academic program

coordinator column is DESIGNED: the relationship 'coordinator of the program' is not implemented yet. The other rows describe the target policy for the modules of Evaluation 3.

| Asset | Why it matters |
|---|---|
| Student academic profile and readiness evidence references | Privacy of students (QR-01) |
| Restricted evaluation comments and sensitive incidents | Dignity and safety (QR-02) |
| Placement status and audit history | Integrity of the process (QR-03, AD-01) |
| Rule configuration | Integrity of the gates (AD-04) |

| ID | Boundary | What crosses and how we treat it |
|---|---|---|
| TB-1 | Browser / client -> API | Untrusted input crosses here. Identity is a header in development (DA-04); real authentication is not designed yet. |
| TB-2 | API -> institutional services (identity, student, employer) | Their answers are data, not truth about our rules; they may fail or be stale; adapters translate and fail closed where needed. |
| TB-3 | API -> database and rule files | Trusted zone; access only with the application credentials; rule files are reviewed in version control. |
| TB-4 | Worker -> notification channel | Messages leave the system; they must contain no restricted data. |

A small threat model is better than 'we use HTTPS'. We list the threats we do not mitigate as well as the ones we do; the identity header (T-01) is acceptable for a course project and not for a pilot. This is a scoped review, not a penetration test.

| ID | Category | Threat | Control | Status | Evidence |
|---|---|---|---|---|---|
| T-01 | Spoofing | A caller sends any X-Mock-User-Id and acts as a coordinator. | Development-only mechanism (DA-04). Production needs real authentication; ports isolate the change. Risk accepted for the course and NOT acceptable for a pilot. | DESIGNED (open risk AR-02) | ADR-007; technical-assumptions.md |
| T-02 | Information disclosure | Enumerating placement ids reveals which exist. | Unrelated users get 404, identical to a missing placement. | VERIFIED | test_unrelated_user_gets_not_found |
| T-03 | Information disclosure | Host organization or student reads RESTRICTED evidence references (agreement, insurance). | Audience projection removes evidence_reference outside the office. | VERIFIED | test_host_organization_sees_readiness_without_restricted_evidence |
| T-04 | Information disclosure | A notification carries restricted data to an external recipient. | Outbox payload contains only placement id and template. | VERIFIED | test_qs06_outbox_payload_has_no_restricted_ |

| ID | Category | Threat | Control | Status | Evidence |
|---|---|---|---|---|---|
| | | | | | data |
| T-05 | Tampering | Someone edits or deletes audit history. | No update/delete interface; application never exposes one. A database administrator could still alter rows: not prevented. | VERIFIED (interface) / open (DB-level) | test_audit_history_can_only_be_appended_and_read; AR-05 |
| T-06 | Tampering | A weaker rule version is published to let a placement start. | Rules are files in version control reviewed by the team; version is stored in each decision. No runtime admin screen yet. | DESIGNED | ADR-004; BL-07 |
| T-07 | Repudiation | A coordinator denies having activated a placement. | Audit entry with actor, time, previous and new value, rule version and correlation id. | VERIFIED | test_ready_placement_is_activated_audited_and_notified |
| T-08 | Elevation of privilege | A student or external user calls the activation endpoint. | Access policy allows only the coordinator staff role; denial is audited. | VERIFIED | test_student_cannot_activate_and_denial_is_audited |
| T-09 | Tampering / integrity | Two concurrent commands produce two success records or two sets of notifications. | Conditional update on status; single transaction. | VERIFIED (deterministic interleaving on SQLite) | test_two_concurrent_activations_only_one_commits |
| T-10 | Denial of service | A flood of requests exhausts the application or the directory. | Not addressed: load is small (DA-01) and no rate limiting is designed. | NOT DESIGNED (accepted, AR-06) | - |

# 11. Deployment and operations

Two environments matter: the demonstration environment that runs today on one computer, and the pilot environment that is only designed. Figure 18 at the end of this section shows both, with the trust boundaries.

| Element | Choice | Why | Decision |
|---|---|---|---|
| Language and framework | Python 3.12, FastAPI | Supplied by the starter, known by the team, small and maintainable (C-03, C-06). | ADR-000 |
| Persistence | SQLAlchemy; SQLite locally, PostgreSQL prepared | One relational database with local transactions (AD-01, AD-03). | ADR-002 |
| Rules | Versioned JSON files + Strategy evaluators | Rules change by program without redeploy (AD-04). | ADR-004 |
| Integration | Ports and adapters over the DH-05 mocks | Services are authoritative and fail (AD-05). | ADR-007 |

| Element | Choice | Why | Decision |
|---|---|---|---|
| Side effects | In-process events + outbox table + worker | Notifications must never undo a decision (QR-04). | ADR-008 |
| Concurrency | Conditional update on status, correlation id | One winner per placement (QS-09). | ADR-010 |
| CI | GitHub Actions: ruff, data validation, pytest | Runs on every push to GitHub to keep main green; run #1 (commit 502188d) passed on 2026-10-11. | Section 12 |
| Web client | Not decided | No driver depends on it; decided in Evaluation 3. | - |

## Environmental assumptions

| Concern | Assumption or configuration |
|---|---|
| Runtime | Python 3.12, FastAPI, SQLAlchemy 2; versions recorded in docs/evaluation-2/validation/versions.txt. |
| Database | SQLite file by default; PostgreSQL 16 profile in docker-compose.yml, not yet exercised by the suite (TP-14). |
| Schema | Created by create_all at start-up; no migrations yet (BL-13). |
| Mocks | In-process classes reading mocks/*.json; failure injected with MOCK_EMPLOYER_FAILURE=EMP-503. |
| Identity | Header X-Mock-User-Id with MOCK_IDENTITY_ENABLED=true; development only (DA-04). |
| Worker | Run on demand with scripts/run_worker.py; a scheduler is a deployment decision for the pilot. |
| Data | DH-05 release-1 data pack; manifest hashes verified (placement_types.csv unchanged). |
| Reproducibility | pip install -e '.[dev]'; python scripts/seed_domain.py; pytest. The same commands run in CI (.github/workflows/ci.yml); run #1 on GitHub passed on 2026-10-11. |

## Signals to diagnose the main failures

| Signal | Interpretation | Source | Status |
|---|---|---|---|
| Activation outcomes by code | Separates normal blocking from failures | Audit entries by action (ACTIVATION_REFUSED, PLACEMENT_ACTIVATED, ACCESS_DENIED); query by resource_id | Available through the audit table |
| Oldest PENDING outbox message | Delivery is not progressing | outbox_messages.created_at where status = PENDING | Available (query); no alert yet |
| FAILED messages | Human action needed | outbox_messages.status = FAILED with last_error | Available (query) |
| STATE_CHANGED rate | Real contention between | API responses with code STATE_CHANGED | DESIGNED (no |

| Signal | Interpretation | Source | Status |
|---|---|---|---|
| | coordinators | | metrics endpoint) |
| Directory failures | Activations blocked by an outage | Blocking reason 'could not be verified' in refusals | Available in audit new_value |
| Correlation | Follow one command end to end | correlation_id in audit, outbox and response | Available |

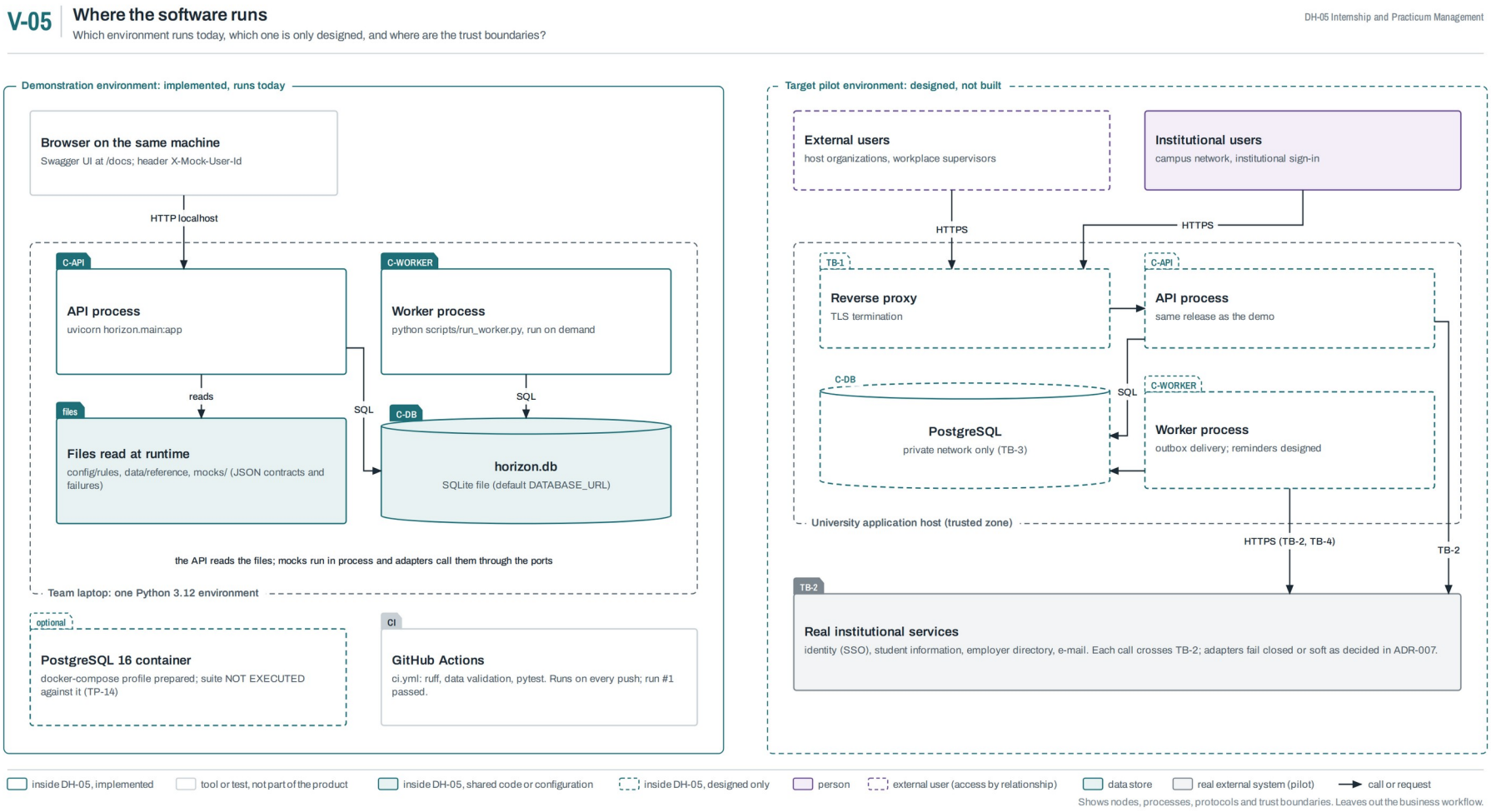



*Figure 18. V-05 Deployment view: the demonstration environment that runs today and the pilot environment that is only designed.*

No container image or shared server exists yet; we do not claim one. The pilot view states what would have to be true before real users touch the system.

# 12. Validation: test strategy, evidence and status

Evaluation 2 does not ask for the working system; that is Evaluation 3. It asks whether the structure is feasible. Our proof of concept attacks the riskiest assumption of the design: that the activation gate cannot be bypassed, even under concurrency, partial failure or a failing directory. It uses the real relational mechanism, not an in-memory substitute.

| ID | Level | Intent | Covers | How | Status |
|---|---|---|---|---|---|
| TP-01 | Unit | Lifecycle transition table and guards | BR-01, BR-03, INV-01..03 | tests/unit/test_placement_lifecycle.py (6 tests) | VERIFIED |
| TP-02 | Unit | Rule evaluators and configuration-only rule change | BR-01, BR-08, QS-07 | tests/unit/test_readiness_rules.py (5 tests) | VERIFIED |
| TP-03 | Unit | Adapters translate payloads and failures; program aliases | AD-05, QS-05 | tests/unit/test_reference_and_adapters.py (7 cases) | VERIFIED |
| TP-04 | Integration | Activation slice end to end through the API | QS-08, BR-01, BR-02 | tests/integration/test_activation_endpoint.py (8 tests) | VERIFIED |
| TP-05 | Integration | Concurrency, atomicity, repetition, correlation, machine-readable codes | QS-09, QS-04, BR-12, INV-04, INV-05, INV-07 | tests/integration/test_concurrency_and_atomicity.py (5 tests) | VERIFIED |
| TP-06 | Integration | Outbox: failed channel, retry, crash, payload content | QS-06, BR-11 | test_outbox.py (1) + test_traceability_evidence.py (3: payload, crash, FAILED after 5) | VERIFIED |
| TP-07 | Integration | Access: unrelated user, wrong role, supervisor relationship, projection | QS-02, QS-01 (partial), BR-09, BR-10 | activation tests + test_traceability_evidence.py (401, 403 host organization, student projection; workplace supervisor at policy level, EU-05) | VERIFIED (slice only) |
| TP-08 | Architecture | Dependency rules, purity of domain, mocks only through M-11, audit append-only | ADR-001, ADR-007, ADR-006 | tests/architecture/test_module_boundaries.py (5 tests) | VERIFIED |
| TP-09 | Data | Seed keeps raw files intact and loads every placement | EU-03, INV-08 | test_traceability_evidence.py (2 tests); scripts/validate_data.py (run locally) | VERIFIED |
| TP-10 | Mutation check | Do the tests detect real defects? Six defects were injected one at a time into a copy of the code (no conditional update, directory failure lets the gate pass, projection leaks evidence, student may activate, forbidden import, worker commits once per run) and each made at least one test fail | Test quality | docs/evaluation-2/validation/mutation_check.py and mutation_output.txt | EXECUTED (6 of 6 detected); script is repeatable, not part of CI |
| TP-11 | Contract | Mock service contracts against the adapters | ADR-007 | Adapter tests use the real mock classes and failure ids; no consumer-driven contract tool | PARTIAL |
| TP-12 | Acceptance | Eligibility, employer decision, academic approval, hours, | QS-01, QS-03, QS-05, | Planned with the modules of | NOT EXECUTED - modules not |

| ID | Level | Intent | Covers | How | Status |
|---|---|---|---|---|---|
| | | evaluation, closure scenarios | BR-05, BR-06 | Evaluation 3 | built |
| TP-13 | Non-functional | Load at the seasonal peak (QS performance is a non-driver) | DA-01 | Not planned unless DA-01 is contradicted | NOT EXECUTED - by decision |
| TP-14 | Environment | Same suite against PostgreSQL; true parallel concurrency | DA-03, QS-09 | Run docker-compose profile with two real connections | NOT EXECUTED - no PostgreSQL in the authoring environment |
| TP-15 | Security | Authentication, rate limiting, dependency scan | T-01, T-10 | Needs real authentication first | NOT EXECUTED - not designed |

## Validation evidence records

Executed on 2026-10-11 with the versions in docs/evaluation-2/validation/versions.txt: 59 tests passed, 0 failed; ruff check reported no issues. The records below separate the expected result from what we observed.

| | |
|---|---|
| **Test** | TP-01 / tests/unit/test_placement_lifecycle.py (6 tests) |
| **Claim** | BR-01, BR-03, INV-01..03: the lifecycle only moves through the transition table; ACTIVE only through activate() with a passed gate. |
| **Status** | PASSED |
| **Setup** | Placement objects built in memory; no database. |
| **Expected** | Activation without a passed gate is refused; other commands cannot reach ACTIVE; closing from PENDING_ACTIVATION is illegal; start date only after activation; interrupted closure allowed from ON_HOLD. |
| **Observed** | As expected (6 passed). |
| **Limitations** | Domain level only; persistence and API are covered by TP-04 and TP-05. |
| **Related** | requirements: US-014; qualities: QS-08; decisions: ADR-003 |
| **Fixture** | None: Placement objects built in memory |
| **Command** | python -m pytest -o addopts="" -v (from the repository root) |
| **Environment** | Windows 11, Python 3.12.1; no database |
| **Independent connections** | not applicable |
| **Executor** | Sebastián Mercado, 2026-10-11 |
| **Executed at** | 2026-10-11 |
| **Trials** | 1 run; the tests are deterministic |

| | |
|---|---|
| **Raw output** | docs/evaluation-2/validation/pytest_output.txt |
| **Source digest** | f7281a5c6c9e7a50 (command: find src tests config -type f \( -name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' \| LC_ALL=C sort \| xargs sha256sum \| sha256sum \| cut -c1-16) |

| | |
|---|---|
| **Test** | TP-02 / tests/unit/test_readiness_rules.py (5 tests) |
| **Claim** | BR-01, BR-02, BR-08, QS-07: pending, expired or missing conditions block; directory failure keeps the gate closed; a rule change is configuration only. |
| **Status** | PASSED |
| **Setup** | Rule evaluators with in-memory conditions; a temporary copy of readiness.json for the configuration test. |
| **Expected** | Each blocking case returns a blocking result; adding CONFIDENTIALITY_FORM for one program changes the result without code changes. |
| **Observed** | As expected (5 passed). |
| **Limitations** | Readiness rules only; eligibility rule types are designed. |
| **Related** | requirements: US-012, US-014; qualities: QS-07, QS-08; decisions: ADR-004 |
| **Fixture** | In-memory conditions; temporary copy of config/rules/readiness.json |
| **Command** | python -m pytest -o addopts="" -v (from the repository root) |
| **Environment** | Windows 11, Python 3.12.1; no database |
| **Independent connections** | not applicable |
| **Executor** | Sebastián Mercado, 2026-10-11 |
| **Executed at** | 2026-10-11 |
| **Trials** | 1 run; the tests are deterministic |
| **Raw output** | docs/evaluation-2/validation/pytest_output.txt |
| **Source digest** | f7281a5c6c9e7a50 (command: find src tests config -type f \( -name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' \| LC_ALL=C sort \| xargs sha256sum \| sha256sum \| cut -c1-16) |

| | |
|---|---|
| **Test** | TP-03 / tests/unit/test_reference_and_adapters.py (7 cases) |
| **Claim** | AD-05, QS-05, EU-03: adapters translate payloads and the three failures; the legacy program code is normalized. |
| **Status** | PASSED |
| **Setup** | The real mock service classes with their documented failure ids (SIS-404, SIS-409, SIS-503). |

| | |
|---|---|
| **Expected** | Failures become port exceptions; read_at is recorded; SWE becomes BSC-SWE; expired agreements are not valid. |
| **Observed** | As expected (7 passed). |
| **Limitations** | Mocks, not the real institutional services; no consumer-driven contract tool. |
| **Related** | requirements: US-012; qualities: QS-05; decisions: ADR-007 |
| **Fixture** | The mock service classes and their JSON contracts in mocks/ |
| **Command** | python -m pytest -o addopts="" -v (from the repository root) |
| **Environment** | Windows 11, Python 3.12.1; no database |
| **Independent connections** | not applicable |
| **Executor** | Sebastián Mercado, 2026-10-11 |
| **Executed at** | 2026-10-11 |
| **Trials** | 1 run; the tests are deterministic |
| **Raw output** | docs/evaluation-2/validation/pytest_output.txt |
| **Source digest** | f7281a5c6c9e7a50 (command: find src tests config -type f \( -name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' \| LC_ALL=C sort \| xargs sha256sum \| sha256sum \| cut -c1-16) |

| | |
|---|---|
| **Test** | TP-04 / test_sc01_*, test_sc02_*, test_directory_down_* |
| **Claim** | QS-08, BR-01, BR-02: activation refused when a condition blocks or the directory is down. |
| **Status** | PASSED |
| **Setup** | Seeded data pack (PLC-2002, PLC-2003); synthetic PLC-9001 with EMP-503 injected. |
| **Expected** | 409 with READY-004 and READY-006; 409 with exactly five blocking results for PLC-2003 (READY-007 expired, READY-008 missing, directory agreement EXPIRED valid until 2026-07-31, ACADEMIC_APPROVAL and WORK_PLAN not registered); 409 'could not be verified'; status unchanged. |
| **Observed** | As expected. |
| **Limitations** | Mocked directory; the real service contract may differ. |
| **Related** | requirements: US-014, US-016; qualities: QS-08; decisions: ADR-003, ADR-004, ADR-007 |
| **Fixture** | DH-05 data pack seeded into a throw-away SQLite file for each test session (tests/conftest.py); fixed date 2026-09-24; synthetic PLC-9001 where stated |
| **Command** | python -m pytest -o addopts="" -v (from the repository root) |
| **Environment** | Windows 11, Python 3.12.1, SQLite 3.43.1 (file in the temp directory), no resource limits, schema |

| | |
|---|---|
| | created with create_all (no migrations, BL-13); versions in versions.txt |
| **Independent connections** | not applicable |
| **Executor** | Sebastián Mercado, 2026-10-11 |
| **Executed at** | 2026-10-11 |
| **Trials** | 1 run; the tests are deterministic |
| **Raw output** | docs/evaluation-2/validation/pytest_output.txt |
| **Source digest** | f7281a5c6c9e7a50 (command: find src tests config -type f \( -name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' \| LC_ALL=C sort \| xargs sha256sum \| sha256sum \| cut -c1-16) |

| | |
|---|---|
| **Test** | TP-05 / test_failure_in_the_middle_rolls_everything_back |
| **Claim** | INV-05: state, audit and outbox commit together or not at all. |
| **Status** | PASSED |
| **Setup** | Event handler writes the outbox rows and then raises. |
| **Expected** | Status PENDING_ACTIVATION, no success audit entry, no outbox rows (checked from a new session). |
| **Observed** | As expected. |
| **Limitations** | Shows nothing was committed; does not distinguish an explicit rollback from a discarded session. |
| **Related** | requirements: US-014; qualities: QS-04, QS-09; decisions: ADR-002, ADR-010 |
| **Fixture** | DH-05 data pack seeded into a throw-away SQLite file for each test session (tests/conftest.py); fixed date 2026-09-24; synthetic PLC-9001 where stated |
| **Command** | python -m pytest -o addopts="" -v (from the repository root) |
| **Environment** | Windows 11, Python 3.12.1, SQLite 3.43.1 (file in the temp directory), no resource limits, schema created with create_all (no migrations, BL-13); versions in versions.txt |
| **Independent connections** | not applicable |
| **Executor** | Sebastián Mercado, 2026-10-11 |
| **Executed at** | 2026-10-11 |
| **Trials** | 1 run; the tests are deterministic |
| **Raw output** | docs/evaluation-2/validation/pytest_output.txt |
| **Source digest** | f7281a5c6c9e7a50 (command: find src tests config -type f \( -name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' \| LC_ALL=C sort \| xargs sha256sum \| sha256sum \| cut -c1-16) |

| | |
|---|---|
| **Test** | TP-05 / test_two_concurrent_activations_only_one_commits |
| **Claim** | QS-09, BR-12, ADR-010: two commands that read the same status cannot both commit. |
| **Status** | PASSED |
| **Setup** | PLC-9001 with every condition complete; two SQLAlchemy sessions on the same SQLite file; the second command runs to completion between the first command's read and write. |
| **Expected** | First writer commits ACTIVE; second raises StaleState; one PLACEMENT_ACTIVATED entry; three outbox rows. |
| **Observed** | As expected (pytest output, docs/evaluation-2/validation/pytest_output.txt). |
| **Limitations** | Deterministic interleaving on SQLite, one trial; not parallel connections on PostgreSQL (TP-14). |
| **Related** | requirements: US-014; qualities: QS-04, QS-09; decisions: ADR-002, ADR-010 |
| **Fixture** | DH-05 data pack seeded into a throw-away SQLite file for each test session (tests/conftest.py); fixed date 2026-09-24; synthetic PLC-9001 where stated |
| **Command** | python -m pytest -o addopts="" -v (from the repository root) |
| **Environment** | Windows 11, Python 3.12.1, SQLite 3.43.1 (file in the temp directory), no resource limits, schema created with create_all (no migrations, BL-13); versions in versions.txt |
| **Independent connections** | 2 SQLAlchemy sessions on the same SQLite file, deterministic interleaving |
| **Executor** | Sebastián Mercado, 2026-10-11 |
| **Executed at** | 2026-10-11 |
| **Trials** | 1 run; the tests are deterministic |
| **Raw output** | docs/evaluation-2/validation/pytest_output.txt |
| **Source digest** | f7281a5c6c9e7a50 (command: find src tests config -type f \( -name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' \| LC_ALL=C sort \| xargs sha256sum \| sha256sum \| cut -c1-16) |

| | |
|---|---|
| **Test** | TP-06 / test_failed_delivery_keeps_activation_and_retries, test_qs06_*, test_outbox_message_is_failed_after_five_attempts |
| **Claim** | QS-06, BR-11, ADR-008: a failed notification never undoes an activation; messages are retried and marked FAILED after 5 attempts; no restricted data in payloads. |
| **Status** | PASSED |
| **Setup** | Activation of PLC-9001, then the worker run with a failing channel, with a crash in the middle and with five failures. |
| **Expected** | Placement stays ACTIVE; messages PENDING then FAILED at 5; sent messages are not resent after a crash; payload has only placement_id and template. |
| **Observed** | As expected. |

| | |
|---|---|
| **Limitations** | At-least-once towards the channel: a crash between send and commit can repeat one message. |
| **Related** | requirements: US-030; qualities: QS-06; decisions: ADR-008 |
| **Fixture** | DH-05 data pack seeded into a throw-away SQLite file for each test session (tests/conftest.py); fixed date 2026-09-24; synthetic PLC-9001 where stated |
| **Command** | python -m pytest -o addopts="" -v (from the repository root) |
| **Environment** | Windows 11, Python 3.12.1, SQLite 3.43.1 (file in the temp directory), no resource limits, schema created with create_all (no migrations, BL-13); versions in versions.txt |
| **Independent connections** | not applicable |
| **Executor** | Sebastián Mercado, 2026-10-11 |
| **Executed at** | 2026-10-11 |
| **Trials** | 1 run; the tests are deterministic |
| **Raw output** | docs/evaluation-2/validation/pytest_output.txt |
| **Source digest** | f7281a5c6c9e7a50 (command: find src tests config -type f \( -name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' \| LC_ALL=C sort \| xargs sha256sum \| sha256sum \| cut -c1-16) |

| | |
|---|---|
| **Test** | TP-07 / access tests in test_activation_endpoint.py and test_traceability_evidence.py |
| **Claim** | QS-01 (partial), QS-02, BR-09, BR-10, ADR-005: relationship-based access, 404 vs 403, audited denials, audience projections. |
| **Status** | PASSED |
| **Setup** | Seeded data pack; identities from mock_identities.json (student, host-organization contact, coordinator). |
| **Expected** | Unrelated user gets 404; related but not allowed gets 403; both audited when the placement exists; non-office audiences never see RESTRICTED evidence references; unknown identity gets 401. |
| **Observed** | As expected. |
| **Limitations** | The identity mock has no workplace supervisors, so that case is verified at policy level only (EU-05). |
| **Related** | requirements: US-027, US-028; qualities: QS-01, QS-02; decisions: ADR-005 |
| **Fixture** | DH-05 data pack seeded into a throw-away SQLite file for each test session (tests/conftest.py); fixed date 2026-09-24; synthetic PLC-9001 where stated |
| **Command** | python -m pytest -o addopts="" -v (from the repository root) |
| **Environment** | Windows 11, Python 3.12.1, SQLite 3.43.1 (file in the temp directory), no resource limits, schema created with create_all (no migrations, BL-13); versions in versions.txt |

| | |
|---|---|
| **Independent connections** | not applicable |
| **Executor** | Sebastián Mercado, 2026-10-11 |
| **Executed at** | 2026-10-11 |
| **Trials** | 1 run; the tests are deterministic |
| **Raw output** | docs/evaluation-2/validation/pytest_output.txt |
| **Source digest** | f7281a5c6c9e7a50 (command: find src tests config -type f \( -name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' \| LC_ALL=C sort \| xargs sha256sum \| sha256sum \| cut -c1-16) |

| | |
|---|---|
| **Test** | TP-08 / tests/architecture/test_module_boundaries.py (5 tests) |
| **Claim** | ADR-001, ADR-006, ADR-007: module dependencies follow the written rules; domain code is framework-free; only M-11 imports the mocks; audit is append-only. |
| **Status** | PASSED |
| **Setup** | Static reading of the import statements of src/horizon. |
| **Expected** | No import outside the allowed table; planned modules hold only __init__.py; AuditRecorder exposes only record() and history(). |
| **Observed** | As expected; mutation M5 (forbidden import) made it fail as intended. |
| **Limitations** | Does not see dynamic imports, router wiring or coupling through shared tables. |
| **Related** | requirements: -; qualities: -; decisions: ADR-001, ADR-006, ADR-007 |
| **Fixture** | None: static reading of the source files |
| **Command** | python -m pytest -o addopts="" -v (from the repository root) |
| **Environment** | Windows 11, Python 3.12.1; no database |
| **Independent connections** | not applicable |
| **Executor** | Sebastián Mercado, 2026-10-11 |
| **Executed at** | 2026-10-11 |
| **Trials** | 1 run; the tests are deterministic |
| **Raw output** | docs/evaluation-2/validation/pytest_output.txt |
| **Source digest** | f7281a5c6c9e7a50 (command: find src tests config -type f \( -name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' \| LC_ALL=C sort \| xargs sha256sum \| sha256sum \| cut -c1-16) |

| | |
|---|---|
| **Test** | TP-09 / test_seed_never_modifies_the_raw_files, |

| | |
|---|---|
| | test_seed_loads_every_placement_of_the_pack_once |
| **Claim** | EU-03, INV-08: the seed never edits the raw exports and loads every placement once. |
| **Status** | PASSED |
| **Setup** | Hashes of data/raw before and after the seed; count of placements loaded. |
| **Expected** | Raw files unchanged; every placement of placements.csv (four) is present after the seed; the primary key prevents duplicates. |
| **Observed** | As expected. |
| **Limitations** | The seed is a one-time load, not a migration tool (BL-13). |
| **Related** | requirements: -; qualities: -; decisions: ADR-000 |
| **Fixture** | DH-05 data pack seeded into a throw-away SQLite file for each test session (tests/conftest.py); fixed date 2026-09-24; synthetic PLC-9001 where stated |
| **Command** | python -m pytest -o addopts="" -v (from the repository root) |
| **Environment** | Windows 11, Python 3.12.1, SQLite 3.43.1 (file in the temp directory), no resource limits, schema created with create_all (no migrations, BL-13); versions in versions.txt |
| **Independent connections** | not applicable |
| **Executor** | Sebastián Mercado, 2026-10-11 |
| **Executed at** | 2026-10-11 |
| **Trials** | 1 run; the tests are deterministic |
| **Raw output** | docs/evaluation-2/validation/pytest_output.txt |
| **Source digest** | f7281a5c6c9e7a50 (command: find src tests config -type f \( -name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' \| LC_ALL=C sort \| xargs sha256sum \| sha256sum \| cut -c1-16) |

| | |
|---|---|
| **Test** | TP-10 / docs/evaluation-2/validation/mutation_check.py |
| **Claim** | The tests detect real defects in the design mechanisms. |
| **Status** | EXECUTED (6 of 6 injected defects detected) |
| **Setup** | Each defect injected alone into a copy of the repository; suite run with PYTHONPATH on the copy. |
| **Expected** | At least one failing test per defect. |
| **Observed** | M1..M6 all DETECTED (mutation_output.txt). |
| **Limitations** | Hand-picked defects, not a full mutation-testing tool. |
| **Related** | requirements: -; qualities: -; decisions: ADR-001, ADR-005, ADR-007, ADR-008, ADR-010 |
| **Fixture** | A temporary copy of the repository for each injected defect |

| | |
|---|---|
| **Command** | python docs/evaluation-2/validation/mutation_check.py . (from the repository root; works on a copy) |
| **Environment** | Windows 11, Python 3.12.1, SQLite 3.43.1 (the full suite runs on each copy) |
| **Independent connections** | not applicable |
| **Executor** | Sebastián Mercado, 2026-10-11 |
| **Executed at** | 2026-10-11 |
| **Trials** | 1 baseline run without defects + 1 run per defect (7 runs) |
| **Raw output** | docs/evaluation-2/validation/mutation_output.txt |
| **Source digest** | f7281a5c6c9e7a50 (command: find src tests config -type f \( -name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' \| LC_ALL=C sort \| xargs sha256sum \| sha256sum \| cut -c1-16) |

What the tests do not prove:

- The concurrency test interleaves two sessions deterministically on SQLite. It proves the conditional update logic, not behavior under true parallel load on PostgreSQL (TP-14).
- The rollback test proves that nothing is committed when a handler fails; it does not distinguish an explicit rollback from the session being discarded.
- The outbox is at-least-once towards the channel; a crash between send and commit can repeat one message.
- Only the activation slice exists in code (M-03, M-04, M-06, M-07, M-08, M-09, M-11). The four business modules M-01, M-02, M-05 and M-10, the closure gate and the web client are designed, not implemented or tested.
- No performance, security-scanning or accessibility testing has been run.
- The CI workflow (.github/workflows/ci.yml) ran once on GitHub (run #1 on commit 502188d, passed), but no pull-request review was exercised: the first version was pushed directly to main.
- The architecture test reads import statements only. It does not see dynamic imports, imports made inside the routers (api.py, the composition points) or coupling through shared database tables.
- QS-02 with a real workplace-supervisor identity is verified at policy level only, because the identity mock has none (EU-05).

## Designed, implemented and verified

| Concern | Design | Implementation | Validation |
|---|---|---|---|
| Activation gate, access, audit, outbox (slice) | Documented | Implemented | Verified (59 tests, 6/6 mutations) |
| Concurrency (conditional update) | ADR-010, SEQ-04 | Implemented | Verified on SQLite; PostgreSQL not executed |
| Module boundaries | V-03, V-DEV, dependency rules | Enforced by test | Verified |
| Opportunities, applications, eligibility | Modules, rules, contracts | Not implemented | Not executed |
| Hours, evaluations, incidents, closure gate | Modules, transitions, BR-06 | Not implemented | Not executed |
| Web client and UX | V-UX-01, V-UX-02 | Not implemented | Not executed |
| Pilot deployment, real identity, rate limits | V-05 target, T-01, T-10 | Not implemented | Not executed |

## Implementation foundation

The repository is the Horizon starter adapted by ADR-000, with the DH-05 data pack loaded, the module structure of section 6, a CI workflow and one vertical slice working end to end: the activation gate. It exercises four of the five drivers at once (gate, access, audit, external service). Acceptance criteria of the slice, all automated:

- GIVEN placement PLC-2002 with ACADEMIC_APPROVAL (READY-004) and WORK_PLAN (READY-006) pending WHEN the coordinator activates it THEN the activation is refused, the response names READY-004 and READY-006, the status stays PENDING_ACTIVATION AND an audit entry ACTIVATION_REFUSED is written.
- GIVEN placement PLC-2003 with VALID_AGREEMENT expired (READY-007) and INSURANCE_REFERENCE missing (READY-008) WHEN the coordinator activates it THEN it is refused naming both conditions.
- GIVEN a placement whose required conditions are complete and valid and whose organization agreement is ACTIVE WHEN the coordinator activates it THEN the status becomes ACTIVE, an audit entry with previous and new value is written AND a PlacementActivated notification is placed in the outbox.
- GIVEN the Employer Directory answers 503 WHEN the coordinator activates THEN the activation is refused and nothing changes (fail closed).
- GIVEN the notification channel fails WHEN a placement is activated THEN the activation remains committed and the message stays pending for retry.
- GIVEN a student or an external user WHEN they try to activate a placement THEN access is denied and the denial is audited.

The seed script documents every transformation from the raw exports: 'Blocked' is not stored, HU-STU-5006's legacy code SWE becomes BSC-SWE, and INT-APP-1005 says academic approval APPROVED while the checklist of PLC-2003 has no ACADEMIC_APPROVAL row. We report that inconsistency instead of inventing the missing row.

Definition of Done agreed by the team:

- The story's acceptance criteria (Given/When/Then) pass as automated tests, including at least one failure case (missing data, unauthorized user or failed dependency).
- Code merged into main through a pull request reviewed by another team member.
- CI is green: ruff and pytest.
- Access to the new endpoint is declared in the Access module and covered by at least one denial test.
- Every state change writes an audit entry (checked by test).
- Affected documents updated: ADR (new or superseded), module interfaces, traceability.
- No hard-coded program or placement-type rule; rules go to configuration.
- Demonstrable with the DH-05 data pack.

Status for the activation slice: every item is met except merging through a reviewed pull request, because the first version was pushed directly to main. CI ran on GitHub and passed (run #1, commit 502188d), and the same commands were run locally; their output is in the validation evidence.

# 13. Risks, backlog and plan

Architecture risks are about the structure we chose, not the project risks of Evaluation 1. Owners: architecture owner Sebastián Mercado, backend owner Juan Arbeláez, quality owner Cher Jiménez.

| ID | Risk | P | I | Mitigation or trigger | Owner role | Status |
|---|---|---|---|---|---|---|
| AR-01 | A module imports another module's internals and the boundaries decay. | M | H | Architecture test fails the test suite (and the CI build on GitHub); keep ALLOWED table in sync with the Module Design. | Architecture owner | Mitigated (TP-08) |
| AR-02 | The mock identity header reaches a real pilot as the only authentication. | M | H | No pilot before IdentityAdapter is replaced; decide with IT Services (identity integration is still an open question). | Architecture owner | Open |
| AR-03 | Readiness matrix (CL-020) differs from our union of observed conditions. | H | M | Rules are configuration; replace file and version. Ask the Legal Representative. | Backend owner | Open (EU-02) |
| AR-04 | PostgreSQL behaves differently from SQLite for conditional update or JSON columns. | L | M | Run the suite on PostgreSQL before Evaluation 3 (TP-14). | Quality owner | Open |
| AR-05 | Audit rows altered directly in the database. | L | H | Restrict database permissions in the shared environment; consider hash chaining only if an | Architecture owner | Open |

| ID | Risk | P | I | Mitigation or trigger | Owner role | Status |
|---|---|---|---|---|---|---|
| | | | | audit requires it. | | |
| AR-06 | No rate limiting or abuse protection. | L | L | Accepted at this load (DA-01); revisit with a real pilot. | Backend owner | Accepted |
| AR-07 | Team overload: four business modules (M-01, M-02, M-05, M-10), the closure gate and the web client remain for Evaluation 3. | H | H | Vertical slices in the order BL-01 to BL-05; stop adding scope. | Team | Open |
| AR-08 | CL-018 stays open and the access policy for restricted fields is wrong. | M | H | Deny by default; the policy lives in one module. | Backend owner | Open |

| ID | Item | Why now | Depends on | Stories | Owner role |
|---|---|---|---|---|---|
| BL-01 | Applications and eligibility slice (M-02 + M-06 rules + StudentInformationAdapter in use) | Second most important rule set; uses the SIS failure policy (QS-05) | - | US-005, US-007 | Backend owner |
| BL-02 | Academic approval and work plan versions; events complete readiness conditions | Feeds the gate with real data instead of seeded rows | BL-01 | US-010, US-011 | Backend owner |
| BL-03 | Hours in three levels and review | Next highest-risk data (disputes) | BL-02 | US-017, US-018, US-019 | Backend owner |
| BL-04 | Evaluations, incidents and restricted projections | QS-03; closes the privacy scenarios | BL-03 | US-020, US-022 | Backend owner |
| BL-05 | Closure gate and CompletionEvidencePort | BR-06 | BL-03, BL-04 | US-024, US-025 | Backend owner |
| BL-06 | Negative tests for invalid rule configuration and directory 404/409 on the activation path; a workplace-supervisor fixture identity for QS-02 end to end | Closes FM-03, FM-04, FM-14, EU-05 | - | - | Quality owner |
| BL-07 | Rule publication: who may change a rule version and how it is reviewed | T-06; CL-004, CL-020 | Client answer | - | Architecture owner |
| BL-08 | API versioning convention and error format ADR | Contract stability | - | - | Architecture owner |
| BL-09 | Run suite on PostgreSQL with parallel connections | TP-14, AR-04 | - | - | Quality owner |
| BL-10 | Web client technology decision | Evaluation 3 UI | BL-01 | - | Team |
| BL-11 | Add eventId and schemaVersion to domain events | Event contract before any extraction | - | - | Architecture owner |
| BL-12 | Outbox archive and retention policy | Data lifecycle | - | US-030 | Backend owner |
| BL-13 | Versioned schema migrations | create_all cannot change an existing schema | BL-09 | - | Backend owner |

| When | Work | Responsible |
|---|---|---|
| By 2026-10-11 (Evaluation 2 delivery) | Submit the Evaluation 2 package (architecture document, ADRs, workbook and repository). Every member rehearses the trace of one requirement in both directions. | Sebastián Mercado, Juan Arbeláez, Cher Jiménez |
| 2026-10-12 to 2026-10-21 | Evaluation 3, first iteration: BL-01 applications and eligibility; BL-06 negative tests; BL-08 error format ADR; BL-09 suite on PostgreSQL. | Juan Arbeláez (backend), Cher Jiménez (quality), Sebastián Mercado (BL-08) |
| 2026-10-22 to 2026-10-31 | Evaluation 3, second iteration: BL-02 approvals and work plans; BL-03 hours; BL-10 web client decision. | Juan Arbeláez; team for BL-10, led by Sebastián Mercado |
| 2026-11-01 to 2026-11-08 | Evaluation 3, third iteration: BL-04 evaluations and incidents; BL-05 closure gate; demo of one placement end to end; update ADRs and views. | Juan Arbeláez, Cher Jiménez, Sebastián Mercado |
| 2026-11-08 (Evaluation 3) | Next architecture review point: does the structure survive execution? Every change goes to a new or superseding ADR. | Sebastián Mercado (architecture owner) |
| 2026-11-29 (Evaluation 4) | Final delivery: remaining backlog items, final evidence and project finished. | Sebastián Mercado, Juan Arbeláez, Cher Jiménez |

# 14. Traceability

Every arrow must be defensible, in both directions. A matrix is the index of the argument, not a substitute for it.

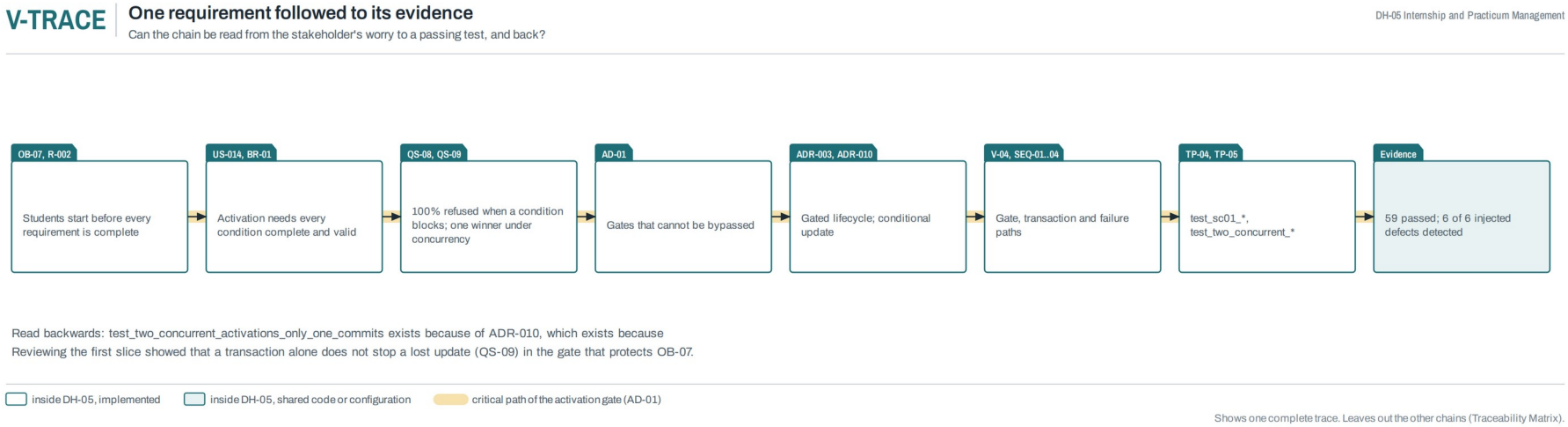



*Figure 19. V-TRACE One requirement followed from the stakeholder's worry to the evidence.*

| Requirement | Rules | Drivers / decisions | Design artifacts | Tests |
|---|---|---|---|---|
| CAP-04 Readiness gate | BR-01, BR-02, BR-03, BR-04, BR-12 | AD-01, AD-05; ADR-003, ADR-004, ADR-007, ADR-010 | V-04, V-TXN, SEQ-01..04, lifecycle; ActivatePlacement, Readiness Gate | TP-01, TP-02, TP-04, TP-05 |
| US-027, US-028 Access | BR-09, BR-10 | AD-02; ADR-005 | V-03, access matrix, threats T-02, T-03, T-08; AccessPolicy | TP-07 |
| CAP-06 History | BR-07, BR-08 | AD-03; ADR-006, ADR-010 | V-DATA, V-STATUS; Audit Recorder | TP-05, TP-08 (append-only), test_qs04_* |
| US-030 Notifications | BR-11 | AD-05; ADR-008 | SEQ-05, V-02; outbox, worker | TP-06 |
| CAP-02 Eligibility | BR-08 | AD-04, AD-05; ADR-004, ADR-007 | M-02, M-06, StudentInformation Port | TP-03 (adapter); TP-12 NOT EXECUTED |
| CAP-03 Three approvals | BR-05 | AD-03; ADR-006 | M-02, V-DOM | TP-12 NOT EXECUTED |
| CAP-05 Verified hours | BR-06 | AD-03, AD-04; ADR-004, ADR-006 | M-05, V-DOM | TP-12 NOT EXECUTED |
| CAP-06 Closure | BR-06 | AD-01; ADR-003 | lifecycle, closure gate | TP-12 NOT EXECUTED |
| CAP-01 Publication | - | ADR-001, ADR-006 | M-01 | TP-12 NOT EXECUTED |

| Driver | Scenarios | Evidence | Decisions | Modules / patterns | Verification |
|---|---|---|---|---|---|
| AD-01 | QS-08, QS-09 | OB-07, OB-11, ASM-006, R-002, R-007 | ADR-001, ADR-002, ADR-003, ADR-010 | M-03, M-04, M-05; Facade (P-06) | SC-05, SC-12; test_activation_endpoint.py (test_sc01_*, test_sc02_*, test_directory_down_*); test_concurrency_and_atomicity.py |
| AD-02 | QS-01, QS-02, QS-03 | CL-009, CL-018, ASM-007, R-003, R-004 | ADR-005 | M-07 (used by all); - | SC-07, SC-08, SC-09; test_student_cannot_activate_*, test_unrelated_user_*, test_host_organization_sees_*, test_qs02_* |
| AD-03 | QS-04, QS-09 | OB-12, F-02, CL-019 | ADR-002, ADR-006, ADR-010 | M-08 (used by all); - | SC-10; audit assertions in test_activation_endpoint.py; test_one_correlation_id_*; test_qs04_* |
| AD-04 | QS-07 | F-24, CL-004, CL-020, ASM-004, R-001, EU-01, EU-02 | ADR-004 | M-06; Strategy (P-01), Factory Method (P-02) | SC-03, SC-05; tests/unit/test_readiness_rules.py::test_rule_change_is_configuration_only |
| AD-05 | QS-05, QS-06, QS-08 | F-17, F-18, ASM-002, ASM-003, ASM-009, R-005 | ADR-007, ADR-008 | M-09, M-11; Adapter (P-03), Observer (P-04) | SC-11; tests/unit/test_reference_and_adapters.py, tests/integration/test_outbox.py, test_qs06_* |

| Story | Module | Driver | Decision | Component / pattern | Success |
|---|---|---|---|---|---|
| US-001 | M-01 | - | ADR-001 | OpportunityService | SC-01 |
| US-002 | M-01 | AD-04 | ADR-004 | OpportunityService | SC-01 |
| US-003 | M-01 | AD-03 | ADR-006 | OpportunityService, audit | SC-01 |
| US-004 | M-01 | AD-02 | ADR-005 | Search + projection | SC-02 |
| US-005 | M-02 | AD-05 | ADR-007 | ApplicationService | SC-02 |
| US-006 | M-02 | - | ADR-001 | ApplicationService | SC-02 |
| US-007 | M-02, M-06 | AD-04, AD-05 | ADR-004, ADR-007 | Strategy (P-01), Adapter (P-03) | SC-03 |
| US-008 | M-02 | AD-02 | ADR-005 | Projection | SC-02 |
| US-009 | M-02 | AD-03 | ADR-006 | ApplicationService, events | SC-04 |
| US-010 | M-03 | AD-03 | ADR-006 | WorkPlan versions | SC-04 |
| US-011 | M-02 | AD-01 | ADR-003 | Three decision records | SC-04 |
| US-012 | M-04, M-06 | AD-04 | ADR-004 | Readiness checklist, Strategy | SC-05 |
| US-013 | M-04 | AD-03 | ADR-006 | Evidence references | SC-05 |
| US-014 | M-03, M-04 | AD-01, AD-05 | ADR-003, ADR-007 | ActivationGate, EmployerDirectoryPort | SC-05 |
| US-015 | M-03 | AD-02, AD-03 | ADR-005, ADR-006 | Supervisor assignment (relationships) | SC-05 |
| US-016 | M-03 | AD-01 | ADR-003 | Lifecycle transition table | SC-05 |
| US-017 | M-05 | AD-03 | ADR-006 | HourEntry | SC-06 |
| US-018 | M-05 | AD-03 | ADR-006 | Hour review records | SC-06 |
| US-019 | M-05 | AD-03 | ADR-006 | Dispute resolution | SC-06 |
| US-020 | M-05, M-07 | AD-02 | ADR-005 | Incident visibility | SC-09 |
| US-021 | M-03 | AD-03 | ADR-006 | WorkPlan versions | SC-10 |
| US-022 | M-05, M-07 | AD-02 | ADR-005 | Evaluation projections | SC-07 |
| US-023 | M-05 | - | ADR-001 | Reflection record | SC-07 |
| US-024 | M-05, M-06 | AD-01, AD-04 | ADR-003, ADR-004 | Closure gate, Strategy | SC-12 |
| US-025 | M-03 | AD-01 | ADR-003 | Lifecycle transition table | SC-12 |
| US-026 | M-08 | AD-03 | ADR-006 | Append-only audit | SC-10 |
| US-027 | M-07, M-11 | AD-02, AD-05 | ADR-005, ADR-007 | IdentityAdapter | SC-11 |
| US-028 | M-07 | AD-02 | ADR-005 | Access policy + projections | SC-08, SC-09 |
| US-029 | M-10 | - | ADR-002 | Read-only queries | SC-02 |
| US-030 | M-09 | AD-05 | ADR-008 | Observer (P-04), outbox | SC-11 |

### Reverse traces

**Start from the worker (C-WORKER).** It exists because a failed notification must never cancel an activation (US-030, BR-11, QR-04, QS-06). ADR-008 chose a transactional outbox. The worker reads PENDING rows from C-DB, calls the channel and commits per message without ever touching a placement. It appears in V-02, V-05 and SEQ-05, and TP-06 challenges it. A worker without the outbox table or without the crash test would be a broken reverse trace.

**Start from the conditional update in SqlPlacementRepository.save.** It exists because two coordinators can act on the same placement (QS-09, BR-12). ADR-010 chose it over row locks. It is not an authorization mechanism and does not replace the gate; it only guarantees one winner. SEQ-04 shows it and test_two_concurrent_activations_only_one_commits verifies it on SQLite.

# 15. What is not decided yet

Eleven clarifications of Evaluation 1 are still open; seven affect the architecture. Each unknown sits behind configuration or behind one module, so the client's answer changes one place.

| Clarification | Still unknown | How the design stays safe meanwhile | Decision |
|---|---|---|---|
| CL-008 | Who may authorize an activation with a missing condition | The gate supports an exception only as a separate, audited command; not implemented until the role is confirmed. | ADR-003 |
| CL-017 | Who resolves disputed hours | The resolving role is a configuration value (default: experiential-learning coordinator, US-019). | ADR-004 |
| CL-018 | Who may read restricted comments and sensitive incidents | Deny by default except the office, the faculty supervisor of the placement and the author of the content (a workplace supervisor reads their own comments); policy lives only in M-07. | ADR-005 |
| CL-020 | Mandatory readiness conditions per placement type | Initial configuration from observed data (EU-02), versioned, replaceable without code. | ADR-004 |
| CL-004 | Exact eligibility rules per program | Rule types derived from eligibility_reviews.csv; parameters per program in configuration. | ADR-004 |
| CL-016 | Whether the reflection is mandatory to close | Completion requirement configurable per program. | ADR-004 |
| CL-001 | Placement types in the first release (ten defined, five used by opportunities, three by placements) | Placement types are reference data; rules are configured per type; the demo uses the types present in the scenarios. | ADR-004 |

# 16. How this evaluation builds on Evaluation 1

Nothing from Evaluation 1 was discarded or rewritten. Its artifacts remain the reference for the problem, and this document uses them as inputs: the quality requirements became the measurable scenarios of section 3, the risks and clarifications became drivers and open questions (sections 4 and 15), the process stages became the module boundaries (section 6), and the stories and success criteria became the traceability and the tests (sections 12 and 14). Where the data pack or the starter kit added detail, we recorded it as an evidence update in section 1 instead of editing the earlier delivery.

# 17. Evidence index

For every important claim: where to look and how strong the evidence is. The repository README is the entry point and lists the same index with paths.

| Claim | Where to look | Status |
|---|---|---|
| The structure is a modular monolith with 11 modules | Architecture Document s.6-7; ADR-001; V-02, V-03 | DESIGNED + boundaries VERIFIED (TP-08) |
| Dependencies between modules follow the written rules | tests/architecture/test_module_ boundaries.py | VERIFIED (5 tests) |
| The activation gate cannot be bypassed and names every blocking condition | tests/integration/test_activation_endpoint. py; tests/unit/test_placement_lifecycle.py | VERIFIED |
| Two concurrent activations produce one success | tests/integration/test_concurrency_and_ atomicity.py | VERIFIED on SQLite (interleaving); parallel PostgreSQL NOT EXECUTED |
| A failure in the middle of a command leaves no partial data | test_failure_in_the_middle_rolls_ everything_back | VERIFIED |
| A notification failure never undoes an activation | tests/integration/test_outbox.py; test_ qs06_* | VERIFIED |
| Restricted evidence is hidden from non-office audiences | test_host_organization_sees_readiness_ without_restricted_evidence | VERIFIED (evidence references only) |
| Evaluation and incident projections hide restricted fields | Access Policy sheet; ADR-005 | DESIGNED (module M-05 not built) |
| Rules can change without code changes | tests/unit/test_readiness_rules.py::test_ rule_change_is_configuration_only; config/ rules/readiness.json | VERIFIED for readiness rules; eligibility rules DESIGNED |
| The tests detect real defects | docs/evaluation-2/validation/mutation_ output.txt | EXECUTED: 6 of 6 injected defects detected |
| The suite passes and the code is lint-clean | docs/evaluation-2/validation/pytest_ | EXECUTED on the delivery |

| Claim | Where to look | Status |
|---|---|---|
| | output.txt, ruff_output.txt, versions.txt | date (see Validation evidence) |
| The data pack loads and validates | scripts/validate_data.py; data/generated/ validation_report.json | EXECUTED: 120 records accepted, 0 errors, 25 warnings |
| Every transformation of the raw exports is recorded | data/generated/transformation_log.md (written by scripts/seed_domain.py) | IMPLEMENTED; raw files unchanged VERIFIED |
| Traceability in both directions | Traceability Matrix sheet; Architecture Document s.14 | Reviewed by the authors before delivery; not automated in the repository |
| Closure gate, eligibility, hours, evaluations | Module Design; Backlog BL-01..BL-05 | DESIGNED only |
| Behavior on PostgreSQL, load, authentication | Test Plan TP-13..TP-15 | NOT EXECUTED |

# 18. Glossary

| Term | Meaning in this document |
|---|---|
| Architecture driver | A requirement, scenario or constraint whose change would change the structure. |
| ADR | Architecture Decision Record: context, decision, alternatives, consequences, verification, revisit triggers. |
| Audience projection | The version of a resource that one kind of user may see, with forbidden fields removed. |
| Conditional update | A write that succeeds only if the stored value still equals the value that was read (UPDATE … WHERE status = the status read). Also called compare-and-set or optimistic concurrency. |
| Composition point | The place where modules, adapters and mocks are wired together; in this project the routers (modules/*/api.py). |
| Shared kernel | Small shared code that every module may use: the ports and the domain events (horizon/ shared). |
| Fail closed / fail soft | Refuse the operation when a dependency fails / keep the request and finish it later. |
| Modular monolith | One deployable application with explicit internal module boundaries. |
| Port / adapter | An interface owned by the core / the code that implements it for one external system. |
| Transactional outbox | Messages written in the same transaction as the decision and delivered later. |
| Readiness gate | The evaluation of every required condition before activation. |
| Rule version | The label of the rule configuration used for a decision. |
| Invariant | A property that every committed state must keep. |

| Term | Meaning in this document |
|---|---|
| | |
| Evidence | The recorded result of something we actually ran; not a diagram or an expected result. |

# 19. References


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7. Team DH-05. Evaluation 1 deliverables: Product Vision and Scope; Process Analysis AS-IS and TO-BE; Artifacts project discussion; Artifacts project logs and requirements.
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